M-INSTR spec: per-instrument profile layer (InstrumentProfile)

Rev 2.5, 2026-11-04 (ET). APPROVED for implementation; M0–M4 merged. Rev 2.5 re-bases the spec on merged `origin/main` @ **`d3c0c1a`** (merge of PR #98, PR-M4, final head `2418052465215c0795954aae03d4173b6959e20e`; first-parent merges #89 `6853398`, #90 `d2b41b7`, #91 `670e28d`, #92 `63cdff0`, #93 `ecc5436`, #95 `7ec8c95`, #96 `4500c67`, #94 `ed3e4a5`, #97 `5c9ccd2`, #98 `d3c0c1a`). Every cite marked **@d3c0c1a** was read on that commit; `_InstrumentProfile.Function.CS` is abbreviated **IP**, `_CustomSignal.Strategy.CS` **CS**, `tests/InstrumentTests/Program.cs` **IT** (InstrumentTests: 756 passed, 0 failed on the M4 head, .NET 8, box; full check: **23 suites, 4528 passed, 0 failed**). Where the code and this spec differ, the difference is flagged **DIVERGENCE** (§12.5). Older cites below are kept as history and say which commit they were read on. **A1 (Ettienne, 2026-11-04):** keep the strict version — stop any strategy on any bad stream; M4 enforces every refusal globally at StartCalc on every data stream for all `_CustomSignal`-based studies (§11.1 item 1, §12.4). **A2 (Ettienne, 2026-11-04):** `_TestPortMM.Signal.CS` deleted in M4; live-check-sheet Check 7 MM-side needs a replacement MM test signal before that check can run (§12.4). **A3 OPEN:** composite / feed exchange names (e.g. NYSEARCA, NASDAQGS, CMEGLOBEX, AMEX, BATS, CFE) are unmapped pending a live check of real `Bars.Info.Exchange` strings on dxFeed, plus the GMT chart VERIFY (§6.1, §12.4). Pins kept: 6E Default `55576df6a8a99c05`, ES Default `6cff053d1b215324`, AAPL `41886799d411dd0d`. History of the rev 2.4 header follows. **Rev 2.4, 2026-11-04 (ET). APPROVED for implementation; M0–M3 merged.** Rev 2.4 re-bases the spec on merged `origin/main` @ **`5c9ccd2`** (merge of PR #97, PR-M3; first-parent merges #89 `6853398`, #90 `d2b41b7`, #91 `670e28d`, #92 `63cdff0`, #93 `ecc5436`, #95 `7ec8c95`, #96 `4500c67`, #94 `ed3e4a5`, #97 `5c9ccd2`). Every cite marked **@5c9ccd2** was read on that commit; `_InstrumentProfile.Function.CS` is abbreviated **IP**, `_CustomSignal.Strategy.CS` **CS**, `tests/InstrumentTests/Program.cs` **IT** (InstrumentTests: 668 passed, 0 failed on `5c9ccd2`, .NET 8, box). Where the code and this spec differ, the difference is flagged **DIVERGENCE** (§12.5). Older cites below are kept as history and say which commit they were read on. Cost keying (§12.4 D1) was DECIDED by Ettienne on 2026-11-04: costs are keyed by asset class + root (`Future:ES`), implemented in PR-M4. **D2** (profileHash over sessions in exchange time) was DECIDED by Ettienne on 2026-11-04, implemented in PR-M4 (§6.1). **MC verification 2026-11-04 (Ettienne, `5c9ccd2`):** MI-T1 PLEditor compile passed, `eMTPA_SF_Root` / `eMTPA_MCSC_USD` no longer VERIFY; the 6E MC run reproduced its profileHash byte-for-byte on .NET 8; the Portfolio Trader per-symbol test passed (§12.3). History of the rev 2.3 header follows. **Rev 2.3, 2026-11-03 (ET). APPROVED for implementation: all questions decided (Q1–Q10, N1–N6). Rev 2.3 is final and ships with PR-M1b** (#90 commits rev 2.2 byte-identical). Rev 2.2 was based on rev 2.1 as merged in `origin/main` @ `6853398` (PR #89, PR-M0; #88 merged as `cd7e50a`) and adds the PR-M0 dump results (§4.4.2). Data inputs still pending from Ettienne: Q3 IBKR per-side fees for the non-ES/MES roots and the Q7 bid/ask start dates; they do not block PR-M0 to PR-M3. Checked read-only against `github.com/ettiennehugo/MultiCharts.Net` `origin/main` @ `7dbdb1b` (merge of PR #85). PR #88 (“Land #86 + #87 on main”) was OPEN at rev 2.1 and has since merged (`cd7e50a`); its code is quoted where marked “#88 head”, and

`_MtfMath.SymbolRoot` was at L108–119 (doc L108, method L109–119) on `6853398`, i.e. **main before #90**; after #90 (PR-M1a, head `a265f4d`) it is doc L108–119, method **L120–135**. Every file:line below was read on `7dbdb1b` unless marked **VERIFY** or "#88 head". **Prerequisite: PR #88 merges before any M-PR.** #88 changes `_MtfMath` (around `SymbolRoot`, L29–39), `_CustomSignal` L1731–1830 (incl. the same-root check L1776–1785), `_DvpEngine`, the DVP golden fixtures and configHash. After it merges, `_CustomSignal` line numbers from about L1740 shift by up to about 50 lines; each M-PR re-reads its cites on the main it targets. Every M-PR targets `main` and is opened only after the previous one merged (no stacked PRs, §11).

Sources: the repo (every `*.CS`, `specs/*.md`, `tests/*`); `specs/instrument-profile-spec.md` rev 3 (2026-10-24, rev-3 Q1–Q11 DECIDED); `specs/m-dvp-spec.md` rev 6; `specs/m-port-spec.md` rev 2.7; the MultiCharts .NET Programming Guide, "Instrument properties" list (just before §4.2.4 StatusLine; box copy, not in the repo); Ettienne's typed API dump output `_McApiDump3.output.txt` (@ES, 2026-11-01; **box copy only, not in the repo**, and produced by a reflection dump, so it is cited as Ettienne's run output, not as a repo file; PR-M0 commits typed dump output as a fixture); CME contract-spec pages read 2026-11-02 (§8); the box research note `/workspace/m-instr/mc-sessions-research.md` (MC sessions: Programming Guide v1.1 §4.2.2–4.2.3, §4.7.4, MC wiki and forum staff answers, MC-1439; 2026-11-02); Ettienne's rulings (ES $17 and MES $2.50 per round trip, 2026-11-01; sessions from the chart, 2026-11-02; Q3–Q10 accepted, 2026-11-02).

Why a new file. Rev 3 treats "Future = today's ES behaviour" as an identity (IP-4.4) and classifies on `Category.ToString()` substrings (IP-1.1, IP-2.2). The 2026-11-01 audit found the futures path itself is ES-only (CME equity-index session clock, quarterly H/M/U/Z roll, third-Friday expiry, the root parser, the clusters). This spec **supersedes rev 3 where they conflict (§1.2)**; rev 3's DECIDED answers stay in force otherwise. Nothing from rev 3 I1–I6 is coded (`rg 'EInstrumentClass|InstrumentProfile' -g '*.CS'` finds nothing); rev-3 PR-I0 `_InstrProbe` is on main and is retired by PR-M1e.

Scope. M-INSTR is a **base service** in `_CustomSignal`. At `StartCalc()` (and, for session data, on the first `CalcBar` of each stream, §4.5) it: 1. reads every used stream's instrument through the **typed** API (`Bars.Info` / `BarsOfData(n).Info`, `IInstrumentSettings`) and the stream's **chart sessions** (`Bars.Sessions`, `IROList<SessionObject>`); 2. builds one immutable `InstrumentProfile` per stream from detection, the chart's sessions, a per-root **exchange-facts** preset (no times) and the user's **cost** figures; 3. logs detection and the applied configuration; 4. hands each engine its effective settings; 5. throws a clear `EngineNotSupportedException` where an engine cannot apply (one rule, R13).

Tuned parameters (thresholds, bandwidths, gap sizes, degrees) are **not** in profiles (R5).

Not touched. The PriceAction library. No new `[Input]`s. M-PORT behaviour (§9). ES on the current hosts and charts stays **bit-identical**, checked by the `_TestGate0` canary hash, the DVP D9a summary (post-#88 baseline) and the ES fixtures in `tests/*` (§10 MI-T15).

Revision log

Rev	Date (ET)	Change
1	2026-11-02	First draft (62-row audit, rev-3 conflicts C1–C8, Q1–Q10 OPEN

2	2026-11-02	Coder review (37 items, Appendix A) applied, each re-verifie **Sessions come from the chart (Q1/Q2 DECIDED by Ettienn** session/RTH windows in any preset; trading day, day open an derive from typed `Bars.Sessions`; RTH-window presets drop DECIDED (accepted recommendations). One throw-vs-degrad Lint scoped to MC enums with an allow-list. `_InstrProbe` reti PR. Gate0Tests updated with the G0 tick change. ringBars kep aligned with the `willArm` code; cost floor is new opt-in behavi rows (8 added, 5 corrected, counts fixed). F1 widened, F3 sec `pointValue` → `bigPointValue`. M1 split four ways (+ probe reti engine. New questions N1–N6.
2.1	2026-11-03	Ettienne 2026-11-03 09:47 ET: *"1 to 5 is fine, on 6 keep that du directory as the created standard log file"*. N1–N5 DECIDED as N6 DECIDED: `_InstrDump` writes `MCInstrDump_<symbol>_<yyyy <pid>.txt` into the directory of the host's standard `MCSignal*` through the host's existing resolver (§4.4); no new hardcoded becomes approved degraded mode D7. Status APPROVED for implementation.
2.2	2026-11-03	Base: rev 2.1 as merged in `origin/main` @ `6853398` (PR #89). §4 **rewritten** (rev 2.1 steps 3–4 turned ESZ2024 into ESZ2024): le letter run, drop a final month letter followed by a digit when letters; MI-T2 gains ESZ2024, 'ES 12-24', ESZ24:XCME, CLZ202 BRK.B, Z4. **PR-M0 dump results** (Ettienne's 13 `_InstrDump` ru CONFIRMED `Bars.Sessions` filled at StartCalc and identical o (R12 now reads at StartCalc, re-checks on the first bar), sessic follow the chart time zone, per-series sessions, multiple sessi kept, `SessionName` value, 6E priceScale within the 1e-9 tolerar CL/GC tick and big point value. **New MC behaviour:** on the G template `LastBarInSession` did not fire at the 15:15 break, on new **R14** (session ends from the Sessions list end times) and M OPEN (awaiting Ettienne): Portfolio Trader second symbol nc Default session vs session-end bars (QuoteManager zone). B1 (commented out in the build).
2.5.1	2026-11-06	**Ettienne 12:13 ET, verbatim:** *"I approve separate expiry/roll preserving other calendar events, explicitly marking unsuppor expiry status unavailable, and rejecting explicit unsupported e queries."* Own-contract QuarterlyExpiry/RollWeek (`EventConf` default on) are separate from shared calendar rows (Monthly VixExpiry, IndexRebalance, HalfDay, RussellRecon; `EventConfig.sharedCalendar`, default on). Turning one off do other. No inferred per-asset-class event mask and no new `[In` with enforcement on and no roll facts does not inherit equity dates. `Step`/`Classify` keep running and set `EOwnContractStatu` (`isRollWeek` / `isQuarterlyExpiry` are null, not false). An explic query (`ContractRollDate`, `ContractExpiryDate`, `IsRollWeek`, `Se` `InSpan(RollWeek)`) logs Error then throws `unknown-field:rol` `SessionsToExpiry` on `NyseStandIn` still refuses `calendar` (S4) a count. ES/MES/NQ/MNQ and `InstrumentConfig.enabled = f` equity-index quarter. `EventConfig.Canon` gains `shared` and `ow` TableSha, profileHash pins, and the DVP golden configHash a Re-based on merged main `d3c0c1a` (PR #98, PR-M4, final head **DONE:** every §6.3 refusal is Error + throw globally at StartCa stream for all `_CustomSignal`-based studies (**A1**, Ettienne: *"Ke version: stop any strategy on any bad stream"*); the `INSTR ds(` line and the null-profile / `profileHash(…=none)` path are gone Unsupported/unmapped class; invalid `InstrumentConfig` (inc cost keys → `unknown-field:config`); empty root; `tick` (PriceSc

2.5	2026-11-04	0, no preset); non-USD (`currency`); unresolved bar/exchange z `tz:exchange`); missing or changed sessions; failed typed/sessi `mismatch:*`; `calendar` / `months` when rules need them. Still rur no preset (cost/floor Unknown); USD stocks on NYSE/NASDAQ **`exchangeZoneIdOf`** (CS L2587–2615): split the exchange name alphanumerics, exact case-insensitive token match, first mat wins; Chicago CME/XCME/GLOBEX/CBOT/ECBOT/CBT/XCBT/CI NYMEX/COMEX/NYSE/NASDAQ/ARCA; anything else → `tz:exc` Composite names (NYSEARCA, NASDAQGS, CMEGLOBEX, AM unmapped pending live dxFeed strings (**A3 OPEN**) + GMT cha **`InstrumentStreams.Start`:** sets `m_started`/`m_pending` only afte returns (a refused stream is never marked started). **DST test** spring-forward gap, fall-back ambiguous read as standard) k `55576df6a8a99c05`. Pins confirmed in IT: 6E `55576df6a8a99c05` `6cff053d1b215324`, AAPL `41886799d411dd0d`. **A2:** `_TestPortMM.` deleted at Ettienne's request; MI-T14 allow-list entry remove sheet Check 7 MM-side needs a replacement MM test signal (`_RootDump.Strategy.CS` is on main (`e1f5b11`, when `_InstrDum` its four rows stay the MI-T2/MI-T3 fixtures. D1/D2 shipped. Fu **suites, 4528/0**.
2.4	2026-11-04	Re-based on merged main `5c9ccd2` (M0, M1a–e, M2 #96, M3 # merged). **M2 (Ettienne's decisions, #96):** 6E roll = `BusinessDaysBeforeExpiry(5)` (§8.1, replaces "Mon of the exp CL/GC/ZN/6E use `ECalendar.NyseStandIn`, logged on INSTR as `stand-in (CME calendar pending M8)` until PR-M8; a preset w rules and no calendar refuses `calendar` at Build, no weekday anywhere (R6); `expiry-worked.csv` roll column = preset rules dates (MI-T4). **M3 (#97):** reason code `months` (§6.3); RUN suffix `profileHash(ds=h;…;ds=none)` outside configHash (§6.1); log- enforced set = `mismatch:*` + `sessions:changed`, everything els Warning with a null profile (§6.2, two-phase Build); INSTR ex `root(<r> rootSrc=…)` and `dataNotes(…)`, ES and 6E lines pinn pin them; profile numbers printed/hashed with `InstrumentTe` `Num` ("R") for diagnostics only; the profile carries the preset's t the cross-check; §4.3 6E tick text corrected (profile `5E-05`; raw Framework "R" `5.0000000000000009E-05`); pinned profileHash `020deca78d6392b7`, 6E `8e8b3e9b0e230e12`; cost floor only with preset, else `floor(Unknown)` (-1), and a new M4 requirement t keyed by root only (**OPEN decision D1**, §12.4); Portfolio Trad Sessions/ASymbolInfo2 test before M4; `eMTPA_SF_Root` **VERIF** or any dump) until MI-T1/MI-T15; tz labels `unresolved` / `unavailable(<ExceptionType>)`; ClassOf stays in `_CustomSigna` category is a log-only Warning in M3 and must be enforced i `MonthMask.Of` throws without an Error line on static preset da (accepted). §11 statuses with PR numbers; #94 listed as an ex made explicit (FALLBACK_TICK sites, `tickFromMinMove` PriceS floor-Unknown reason, ClassOf enforcement, log-only → enfo `eMTPA_SF_Root`). New §12.4 (open decisions), §12.5 (code-vs-sp Standing rules restated after R14. **D1 DECIDED (Ettienne 20 revision):** costs keyed by asset class + root, key `<Class>:<ROO` `Future:ES` 1700, `Future:MES` 250); a row whose class does not symbol is not used (`cost(Unknown)`, `floor(Unknown src=cost-mismatch)`, one Warning); root-only keying is M3 behaviour; f scope (§5.3, §9, §11.1 item 8, §12.4); configHash unaffected (n includes `InstrumentConfig.Canon()` on `5c9ccd2`); profileHash ES/MES/6E, changes only where a cross-class cost was shown **D2 DECIDED (Ettienne 2026-11-04):** profileHash hashes the s

2.3	2026-11-03	`ChartDaySession` rth) converted to the **exchange** time zone (`exchangeTimeZoneId(Info.Exchange)`, strict lookup), so one i hashes the same on any chart/PT zone; new refusals `tz:bar` / never a silent fallback; INSTR keeps bar-time `sessions(…)` an `xsessions(…)`; expected pins after M4: 6E Default `55576df6a8` zone, ES Default `6cff053d1b215324`; IT pins with exchange-tin unchanged, new tz-invariance tests (§6.1, §11.1 item 9, §12.4, **verification (Ettienne 2026-11-04, `5c9ccd2`):** MI-T1 Compile the one-time import of `_InstrumentProfile.Function.CS`); `eM` `eMTPA_MCSC_USD` compile, VERIFY dropped; `rootSrc=mc` on 6E, T15 6E chart hash `55576df6a8a99c05` reproduced on .NET 8 (r `8e8b3e9b0e230e12` is the synthetic 9-session list); PT per-symb (@6E, @ES, AAPL; ES `e1927ccf1ea6f41e`, AAPL `41886799d411d` item closed; PORT publish lines stamped `0001-01-01` / zone `?` item 10). Prepared after #90 (which commits rev 2.2 byte-identical); **fi PR-M1b**. Header: the `SymbolRoot` L108–119 cite is main **befor** post-#90 location L120–135. §4.3/MI-T2: `MESH26` example repla `MESZ26`/`MESH27` → `MES`, as in the #90 tests. **@6E zone RESOLVE** QuoteManager custom sessions with ET hours entered as CT error, not MC behaviour); Ettienne switched to exchange sess rerun; the 6E fixture is marked superseded. Folded in Coder's `templates-research.md`: the built-in E-mini GLOBEX template halt removal, pre-2012 15:30–16:30 row) and sits on the D9a/ (environment caveat, engines take sessions as given); "Combi sessions per day; templates editable in QuoteManager → Tool Templates; `LastBarInSession` fires on rows with Session End not documented) as R14's rationale, with sources. Portfolio T OPEN. **MC stores the root (credit: Ettienne):** `Bars.Info.ASy` (`MTPA_MCSymbolInfo2`, PG §4.2.3, pg11.txt L901, L3828–3867) b resolver's primary root, with the #90 parser as fallback wher flagged in `UnsupportedFields` (typed); `!SameRoot` → Warning, B1/B2 lead with `SymbolExpiry`/`SymbolCurrency`; new B3 cross- `SymbolCategory`/`MinMove`/`PriceScale`/`BigPointValue` vs `Bars.I` **Resolved by Ettienne `_RootDump` 2026-11-03, relayed by Cod** SymbolRoot/Currency/Category/Exchange/MinMove/PriceSca CONFIRMED; parser = pure fallback on `(UnsupportedFields eMTPA_SF_Root) ≠ 0` or empty root; SymbolExpiry placehold currency from SymbolCurrency (B2); MC-internal check Pric Info.PriceScale = 1 ±1e-9 and BPV equal, else throw; MI-T2/M removed from §12.3. **@6E CLOSED:** rerun `01M6HTKV3SV1Q5QD` (17:00–16:00 Exchange, ends 16:00, matches ES; `bvo(TickCour` instrument/session facts only) is the 6E fixture; `01M6HSDCCJZXJHNB165KSF6Z5B.txt` discarded; 6E rerun remov Setup note: restart MC after changing QuoteManager session

0. Rules (normative; every PR must comply)

- **R1 Typed API only.** Never use MultiCharts enums or API members as strings: no `Enum.Parse`/`TryParse`/`GetNames` on an MC enum, no `<MC enum>.ToString()` used in a comparison, parse or hash, no reflection by member name on MC types. Formatting an MC enum into a human-readable log line is allowed (nothing reads it back). Repo-own enums parsed from protected config are outside R1 (§10 MI-T14 allow-list).

Where a typed member is uncertain it is **VERIFY** with a typed dump line (§4.4); a line that does not compile is the answer (the member does not exist). Never guess.

- **R2 Build hygiene.** New `.CS` → `<Compile Include>` in `PLStudies.csproj` + one-time PowerLanguage Editor import (listed in the PR). New test projects → `PLStudies.Tests.sln`, `CheckForOverflowUnderflow=true`, real-source linking via a `gen.sh` awk region (`// <instr-core>` … `// </instr-core>`), as in `tests/InstrProbeTests`. C# 7.3.
- **R3 PriceAction unchanged.** The profile does not touch `EPriceActionPreset` or the PriceAction library; VPFUT keeps its existing preset call.
- **R4 No new `[Input]`s.** Settings go through protected config objects (`InstrumentConfig`, §5.3), like `EventConfig`, `RegimeConfig`, `DailyConfig`, `DvpConfig`, `PortMmConfig`.
- **R5 Self-calibration over tables.** Presets hold **exchange facts only**: tick, big point value, currency, contract months, expiry rule, roll rule, holiday calendar (where verified), cluster. **No session or RTH times** (Q1/Q2). ES-tuned values (e.g. `RollGapTicks` 8, `KdeBandwidthTicks` 6, `_ProfileStats` 4-tick disagreement) stay as today with a CFG note "ES-tuned default", or become relative to a running statistic in a later PR (§7).
- **R6 Auto-detect, apply, log, refuse.** No silent fallback. `TICK_FALLBACK = 0.01` (`_CustomSignal` L29, `resolveTick` L2287–2301 with a bare `catch {}`) becomes an Error + throw (PR-M1b; **done** in #91: CS `resolveTick` L2347–2368 @5c9ccd2, Error line via `traceAlways` then throw). **No weekday fallback anywhere (rev 2.4, #96):** every expiry/roll computation needs an assigned business-day calendar. `InstrumentCalendars.Require` (IP L450–455) and `InstrumentRules.requireCalendar` (IP L467–471) refuse with reason `calendar`; Build calls `Require` whenever the profile has an expiry or roll rule (IP L803), so a preset (or override row) with roll rules but no calendar (`Observed/None`) is refused at profile build. The old weekday default gave wrong dates (CLF26 12-22, ZNH24 03-20, ZNZ25 12-22; IT L209, L228). In M3 that `calendar` refusal was log-only (§6.2); **M4 enforces it** (A1). Known fallbacks audited and **removed in PR-M4** (§11): `FALLBACK_TICK = 0.01` in `_VolumeProfileBase` / `_VpConfluenceEngine`, and CS `tickFromMinMove` returning `MinMove` when `PriceScale == 0` (and `Math.Abs` of a negative ratio).
- **R7 Volume basis.** Futures (`Future`/`FutureRolover`) on Build Volume On = Tick Count → Error + throw; non-futures on Tick Count are tagged `TickCount`. **Already on main** (`_VolumeProfileBase` L1429–1461, L1505–1506); M-INSTR only moves it into `volumeBasis`.
- **R8 Sizing.** 1% risk per trade; hard max-qty/max-notional caps reduce size (`sizeForRisk`, L2921–2953). The portfolio path (`_PortMM`, deny-not-reduce) is unchanged. Details and the actual code path: §9.
- **R9 Costs.** ES $17.00 and MES $2.50 per round trip (Ettienne). Other roots: method DECIDED (Q3), figures need Ettienne's IBKR per-side fees per root (data input).
- **R10 Logs.** One `File.AppendAllText` per line; skipped entries at **Info**; doubles formatted with `CultureInfo.InvariantCulture`.
- **R11 Facts.** Cite file:line; exchange facts carry a source or **VERIFY**.
- **R12 Sessions come from the chart (DECIDED, Q1/Q2).** Session times, the trading-day boundary and the session end come from the stream's chart sessions via typed `Bars.Sessions` / `SessionObject`. The chart's session template is the source of truth (normally the instrument's MC "Default"). Engines do not second-guess it and do not warn about which template the user chose. Sessions are per stream (each data series has its own, **CONFIRMED** §4.4.2; in Portfolio Trader each symbol has its own, with one global time zone) and are read **in StartCalc and re-**

checked on the first CalcBar of each stream (both reads identical in all 13 dump runs, **CONFIRMED** §4.4.2; a difference throws `sessions:changed`, S1). `Bars.Sessions` exists (it compiles on main) but is documented only on the MC forum; its behaviour is now taken from the PR-M0 dump (§4.4.2), not from the documentation. The template name (`Info.SessionName`, Programming Guide) is logged and never parsed. Holidays cannot be read from code, so holiday calendars stay in the repo as calendars only (PR-M8). The DV/MTF session-mismatch Warning between streams (#86/#88 `checkHtfSessions`), including its MC-1439 / Realtime-History Matching caveat, stays.

- **R13 Throw vs degrade (one rule).** An engine **throws** when it cannot apply to the instrument or to the chart's sessions. It runs **Degraded** only in a mode Ettienne has approved, which still produces correct results from reduced inputs; each such mode is listed in §6.4 with what is reduced. An output that is *undefined* for the given sessions (e.g. an overnight high on a chart with no overnight segment) is NaN, logged once at Info, and is not a degrade; §6.4 lists every such output.
- **R14 Session ends come from the Sessions list (NEW, rev 2.2).** A session ends at the `EndTime` of a `SessionObject` in the stream's Sessions list, not where `Bars.LastBarInSession` happens to be true. Observed (§4.4.2, `01M6HSDBFRQ9ZZFA4NBXZ8QZXC.txt`, `01M6HSDBFH2D6H21PMYXYKFJHV.txt`): on the CME GLOBEX template (sessions 17:00–15:15 and 15:30–16:30 CT) `LastBarInSession` was true only on the 16:00 bar, **not at the 15:15 break**. The session reader and every session-reset engine (VPFUT, AVWAP, FLOW, TOD, RANGE, WEEKLY, DAILY, DVP, G0, REGGATE, ROLL keying) therefore assign each bar to the session containing its close (start exclusive, end inclusive) and detect a session end as: (a) a bar whose close equals the session's `EndTime`, or (b) the first bar that falls in a different session than the previous bar (the end is then attributed to the previous bar), or (c) `LastBarInSession` true (kept only as an extra early signal for the last bar when data stops before the listed end, e.g. ES data ends 16:00 inside a session listed to 16:30). `LastBarInSession` alone is never the only signal. A reset fires **once per (stream, trading day, session index)**: the two-series run logged the same 2026-10-26 16:00 Data1 bar as a session end twice (`01M6HSDBFH2D6H21PMYXYKFJHV.txt`, CalcBar called once per stream update), so resets must be idempotent. Tested by MI-T16. **Rationale (rev 2.3):** each template row has a **Session End** checkbox and MC staff say to tick it on the row that ends the trading day (forum t=45178, https://www.multicharts.com/discussion/viewtopic.php?t=45178); ticking it on every row gave a break at every row (t=5890, https://www.multicharts.com/discussion/viewtopic.php?t=5890). `LastBarInSession` therefore most likely fires only at rows with Session End ticked, which matches the GLOBEX observation (15:15 row presumably unticked). This is **inferred, not documented** (research `/workspace/m-instr/mc-session-templates-research.md` §6), and the tick is not exposed by `SessionObject` (§4.1), so the list `EndTimes` are the only readable boundary; live, the flag can also be late or missed when no tick prints in the last bar (staff, https://www.multicharts.com/discussion/viewtopic.php?t=50358; https://www.multicharts.com/discussion/viewtopic.php?t=48918).

Standing rules (restated, rev 2.4; every PR still checks them): - **No MC enums or API values as strings** (R1): typed members only; MC enum labels are formatted by the host for log lines and never read back (IP header L8–11; lint MI-T14). - **Sessions come from MC session data** (R12): the chart's template (normally Default) via typed `Bars.Sessions`. The only code-side windows are the accepted ES/MES/NQ/MNQ RTH split on a 24-

hour chart (N1, `rthSource=EquityIndexLegacy`, IP `RthSplit` L643–665) and the fixed Asia/Europe/NY clock windows (N5). - **Every new `.cs`** needs a `PLStudies.csproj` `<Compile Include>` and a one-time PowerLanguage Editor import, listed in the PR (R2; `_InstrumentProfile.Function.CS` is csproj L522 @5c9ccd2). - **Error is logged before every throw** that leaves M-INSTR code (`InstrumentError.Fail`, IP L91–98; host `traceAlways`). In M3 a log-only refusal is caught by `InstrumentStreams.Guard` and written once as a Warning instead (§6.2); `MonthMask.Of` is the one accepted exception (§12.4 note). - **No silent fallbacks** (R6), including no weekday business-day fallback.

---

1. Relation to `instrument-profile-spec.md` rev 3

1.1 Kept from rev 3

The engine × class support matrix (rev 3 §8.1) and its DECIDED answers: rev-3 Q1 ETF = Stock; rev-3 Q2 FLOW throws on stocks; rev-3 Q3 VPFUT throws on non-futures; rev-3 Q4 no force flag; rev-3 Q5 DVP/G0 Unsupported on FX, VP Degraded (visual) on FX; rev-3 Q7 `profileHash` outside configHash; rev-3 Q8 `_RegimeEngine` on ES/MES only; rev-3 Q9 ES $17; rev-3 Q10 USD only; rev-3 Q11 EVENT Degraded on USD FX pairs. Also kept: the INSTR/CFG log idea, the `EngineNotSupportedException` format (rev 3 §6), the cost-floor formula (IP-8.3).

1.2 Conflicts (this spec wins)

#	Rev 3	Conflict	Rev 2
C1	IP-1.1/IP-2.2 classify by `Category.ToString()`; probe uses `Enum.GetNames`	R1	Exhaustive `switch` on `ESymbolCategory` in the host (§4.2).
C2	IP-1.4 “Build Volume On is not on `IInstrumentSettings`”	Wrong: `Info.BVO` is typed (`_VolumeProfileBase` L1505)	`volumeBasis` from `Info.BVO`.
C3	IP-4.4 “Future = today identity”	Futures path is ES-only	Per-root exchange facts (§8) + chart sessions; ES bit-identical.
C4	IP-8.1 “q is never reduced”	Hard caps reduce today (L2943–2949)	§9 describes the actual code.
C5	IP-8.3 MES $2.47	R9	MES $2.50, floor 20 ticks (integer-cents formula, §9).
C6	PR-I0 `_InstrProbe` uses reflection (`_InstrProbe` L4/67/74/75/144; `_InstrProbeCore` L5/54/126/167)	R1	Typed `_InstrDump` (PR-M0); `_InstrProbe` + `tests/InstrProbeTests` retired in PR-M1e (Q6 DECIDED).
C7	Rev-3 line numbers	Stale	Re-read on `7dbdb1b`.
	PR-I4 “no futures	Other CME product groups have other	Calendar rule §6.4 S4

C8	holiday re-route”	holidays/early closes	and PR-M8; ES keeps `_CmeHolidays`.
C9	Rev-3 session handling = CME equity-index clock for every future	R12	Sessions from `Bars.Sessions` (§4.5); no session times in presets.

---

2. Purpose

Today an engine run on anything other than @ES/@MES silently uses ES facts, guesses from strings, or fails later with an unclear error. M-INSTR: (1) detects the instrument from typed MC data; (2) takes session times, trading day and session end from the chart’s sessions; (3) resolves the exchange facts each engine needs (tick, big point value, currency, contract months, expiry and roll rule, calendar, cluster) from detection plus a small exchange-facts preset per root; (4) applies them, ES bit-identical; (5) logs; (6) refuses clearly under R13.

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3. Audit: ES- and futures-specific hardcoding on `origin/main` @ `7dbdb1b`

Method: `rg` over every `*.CS`, `specs/*.md`, `tests/*` for 0.25, 12.5, $50, `"ES"`/`@ES`, H/M/U/Z, `ThirdFriday`, `TimeSpan(9,30)`, (16,0), (16,15), (17,0), (18,0), (15,15), Globex, Eastern/Central, USD, `Category.ToString`, `SymbolRoot`, `PortfolioCluster`, cost literals (471 raw hits, `/workspace/m-instr/grep1.txt`), reduced to one row per distinct hardcoding, plus the 8 rows the rev-1 review found missing (63–70). Column “Sess” = **S** where the row is replaced by chart-session data under R12 (§4.5), **T** where it is a frozen pre-registered trial value (kept, coverage-checked, §6.4 S3).

Engine tags: CS host `_CustomSignal`; VPFUT `_VolumeProfileFuturesBase`; CLOCK `_CmeSessionClock`; CAL holiday calendars; ROLL `_RollDetector` + host roll code; EVENT `_EventCalendar`; G0 `_Gate0`/`_TestGate0`; FLOW `_OrderFlow`; REGGATE `_RegimeGate`/`_Test6`; REGIME `_RegimeEngine`; DAILY `_DailyContext`; TOD `_TodNormaliser` (host side); DVP `_DvpEngine`/`_DvpEngineTest`; SIZE CS-4 sizing; COST; PORT M-PORT; MTF `_MtfMath`; LEVEL, AVWAP, RANGE, WEEKLY, PSTATS `_ProfileStats`; LEGACY pre-engine studies.

3.1 Table

#	file:line	What is hardcoded
1	`_CustomSignal` L1603–1618	`PortfolioCluster` switch on ro strings: EQIDX (L1611), RATES (ENERGY (L1613), METALS (L16 others fall through to the raw r
2	`_CustomSignal` L2462	`autoTradingDay(barTime, Bars.Info.Category.ToString(` enum → string (R1)
3	`_CustomSignal` L2581–2587	`autoTradingDay` substrings “FOREX”/“CURRENCY”/“CRYPT(day = ET+7h; crypto = UTC date

4	`_CustomSignal` L2587–2615 @d3c0c1a	`exchangeZoneIdOf` token map (I replaced the substring `Contain` match): split `Info.Exchange` on alphanumerics, exact case-inse token match, **first matching to wins**. Chicago → Central: CME, GLOBEX, CBOT, ECBOT, CBT, XC CBOE. New York → Eastern: NY COMEX, NYSE, NASDAQ, ARCA. Anything else → Build refuses `tz:exchange`. Composite / feed r (NYSEARCA, NASDAQGS, CMEGLOBEX, AMEX, BATS, CFI unmapped pending a live chec real `Bars.Info.Exchange` string dxFeed (**A3 OPEN**, §12.4). (Hist dump @CL/@GC showed 18:00- on Exchange and Local, §4.4.2 NYMEX/COMEX → Eastern entr unchanged.)
5	`_CustomSignal` L29, L2287–2301	`TICK_FALLBACK = 0.01` with a b `catch {}`
6	`_CustomSignal` L2062	`_NyseHolidays` created for ever instrument
7	`_CustomSignal` L3228–3240	`tradingDayCount` uses NYSE for stocks (warns once)
8	`_CustomSignal` L921	`isRthBar` returns true on the ge base
9	`_CustomSignal` L949	`todMinuteOfSession` default; fu override = 18:00 ET (row 18)
10	`_CustomSignal` L1063–1072	DAILY roll-offset warning assu 18:00 ET splice; `Category == Fu` only (misses `FutureRolover`, F2
11	`_CustomSignal` L1801, L1836	`RollGapTicks = 8` (ES-tuned; `_RollDetector` L12–13)
12	`_CustomSignal` L1866, L1881–1882	Roll warning / `m_rollGlobexKey` require `Category == Future`; `isFuturesCategory` (`_VolumeProfileBase` L1440–14 accepts `FutureRolover`
13	`_CustomSignal` L1840–1842	Roll inheritance keyed by `_MtfMath.SymbolRoot` (row 31). cited L1778–1783, which is the HtfStreams same-root throw L1 1785, also a `SymbolRoot` user, fix PR-M1a.)
14	`_CustomSignal` L749, L766	DVP root = `SymbolRoot(...).ToUpperInvar`
15	`_CustomSignal` L2891–2953	`unitNotional` 0 = entry × BigPointValue, implicit USD
16	`_CustomSignal` L2994–3002	`roundTripCostPerContract` = 2 × Commission + Slippage)
17	`_CustomSignal` L2008	AVWAP `sessionDate = tradinç rth = isRthBar`

18	`_VolumeProfileFuturesBase` L489–493	TOD minute = minutes since 18
19	`_VolumeProfileFuturesBase` L52–63	`DEFAULT_*` strings in ET: Asia, E NY, RTH 09:30–16:15, halt 17:0(Globex 18:00/17:00
20	`_VolumeProfileFuturesBase` L66–77	Same defaults as `TimeSpans` (VP
21	`_VolumeProfileFuturesBase` L361	`tradingDayKey` = `GlobexDate` (18
22	`_VolumeProfileFuturesBase` L396–398	`DefaultRthValueEnd` 16:00 ET (v window, D5)
23	`_VolumeProfileFuturesBase` L472–476	`EOwnProfileWindow { MultiChartsProfile = 0, Glob Rth }`
24	`_VolumeProfileFuturesBase` L517; `_CmeSessionClock` L373–379	`isRthBar` = RTH **value** window 16:00] ET via `IsRthValueBarEt`; `m_rthEnd` 16:15 is the session fla the value window
25	`_VolumeProfileFuturesBase` L975–995	Tick-chunk plan per Globex da
26	`_VolumeProfileFuturesBase` L1627, L1635–1671	VPFUT's own `UseCmeHolidayTab` (L1627) and hooks: `IsGlobexCl` L1635, `IsNoRthDay` via NYSE L1 1646, `HaltEt` L1656, `RthCloseEt` `_CmeHolidays.Rules.Get(date)` L1671
27	`_CmeSessionClock` L48–61	"CME equity-index hours" defa ET: RTH, RthValueEnd, halt, Gl open/close, Asia/EU/NY
28	`_CmeSessionClock` L81	Clock's `UseCmeHolidayTable =` (separate from VPFUT's L1627, 26)
29	`_CmeSessionClock` L248–249	`GlobexTradingDayEt(et)` 18:00 open
30	`_CmeHolidays` L13–45, L65	CME equity-index holiday/early rules (13:15 ET L44, Good Frida ET L45, halts L65)
31	`_MtfMath` L29–39	`SymbolRoot`: letters only; drops trailing month letter. "@6E", "6 → ""·"EUR/USD" → "EUR" (F1)
32	`_RollDetector` L23	`gapTicks = 8`
33	`_RollDetector` L43	`RollRuleMondayFrom = 2022-04`
34	`_RollDetector` L46–51, L56–67	`ThirdFriday`, quarterly `SeriesW`
35	`_EventCalendar` L20	Event rows in ET (correct for U macro)
36	`_EventCalendar` L128	Default event classes (no EIA fo
37	`_EventCalendar` L189	`EventRules.RollRuleMondayFro` 04-01
38	`_EventCalendar` L290–303	`ExpiryDate`/`RuleExpiryDate` 3rd `RollDate` Mon before (2022+) e −8
		`nextQuarterlyExpiry`,

39	`_EventCalendar` L1913–1928	`SessionsToExpiry`, `RuleRollDat` quarterly
40	`_Gate0` L328	`KdeBandwidthTicks = 6`
41	`_Gate0` L357	`tick = 0.25` default (host sets i `_TestGate0` L237; `_Gate0` L461 `study.tickSize = tick`)
42	`_Gate0` L474, L614, L1037/1059/1116/1129/1267, L1070	RTH 09:30–16:00 ET, 30-min pe flat 15:55, 09:35
43	`_Gate0` L1185–1186	Doc: “q × widened stop × $50”; `costFloorTicks` only in comme
44	`_Gate0` L1204	`new _DvpEngine(c, "G0KD02", tick, …)`
45	`_TestGate0` L42, L156–158	“CME Globex” template; $2.25/s 0.5 tick/side = $17
46	`_OrderFlow` L124	`TickChunkBoundaryEt = 17:00`
47	`_OrderFlow` L935	`PlanGlobexDays(...)`
48	`_RegimeGate` L45–48	A2-63: rthOpen 09:30, activeEn exit 15:55, rthClose 16:00 ET
49	`_RegimeGate` L55–62	Roll years, digest `fdd63bd8f179` (L60), `tradingDay = "Globex18` (string, L62)
50	`_Test6` L63, L89, L105–106	ET zone string, Globex key, `RollDetection.ToString()` (rep enum) into the hash
51	`_DailyContext` L30, L33	`{252,126,63}`, `hvAnnualisation`
52	`_RegimeEngine` L272	`sqrt(63/252)`
53	`_LevelStudy` L79	`tickSize = 0.25` (host sets L14
54	`_AvwapManager` L153	`tickSize = 0.25` (host sets L19 L195 rejects ≤ 0)
55	`_RangeProfile` L31	`ringBars = 2880` (a **bar** count)
56	`_WeeklyProfiles` L19, L26	Default source `RthSessions`
57	`_PortMM` L388, L740; `_PortfolioRiskManager` L87, L129	USD-only book (typed `eMTPA_MC` correct)
58	`_DvpEngineTest` L213, L230	`root == "ES"`: roll table ES-only
59	`_DeltaDivRevFuturesLELX` L152, L175, L528–564	Globex state, Globex-day trade
60	`_ExitHalfRiskHalfRoll` L107–113	“One tick” = `1 / PriceScale * BigPointValue` (0.5 on ES if Pric = 100: neither a tick nor $/tick)
61	`_EconomicCalendar` L231; `_EhlersFilter` L149; `_StructuralTargetPatterns.Strategy` L1060; `_GenerateNNTrainingData` L354	FX pip = `PointValue × 10 × BigPointValue`
62	`_EpisodicPivotBase` L32–35; `_ParabolicShortSESX` L51–52; `_BreakoutAutoLELX` L42; `_TrendLine` L2755–2756; `_VolumeProfileBaseLog` L206, L426	US-stock 09:30/16:00 ET; `m_tick` `0.25`

63	`_CustomSignal` L1888–1889	`findTz("Eastern Standard Tim` `GlobexTradingDayEt` for the roll any future
64	`_DvpEngineTest` L119–121; `_Test6` L100–102	Host-generated quarterly `RollD` (`for m = 3..12 step 3: RollDates.Add(EventRules.Rol`
65	`_ProfileStats` L47	`ringBars = 1440` (bar count)
66	`_ProfileStats` L575	`disagree` when
67	`_TestGate0` L157	NQ slippage $2.50 in the canary
68	`_MtfMath` L28	`SymbolRoot` doc says returns "@ code returns "ES"
69	`_DvpEngine` L2445–2447	`RootError`: empty == empty is a so any two digit-led roots pass
70	`_ExitHalfRiskHalfRoll` L123	`GenerateStopLoss(m_firstStop` a price distance passed where takes currency (F3b)

Counts. 70 rows; each counted in exactly one area. Rows 61 and 62 group several legacy files each.

Area	Rows	Count
CS host	2, 3, 4, 5, 8, 9, 10, 63	8
VPFUT	18–24	7
CLOCK and CAL	6, 7, 26–30	7
ROLL	11, 12, 13, 32–34	6
EVENT	35–39	5
G0	40–45, 67	7
FLOW	25, 46, 47	3
REGGATE	48–50	3
DAILY and REGIME	51, 52	2
SIZE and COST	15, 16	2
PORT	1, 57	2
DVP	14, 58, 64, 69	4
MTF	31, 68	2
AVWAP	17, 54	2
LEVEL, RANGE, WEEKLY, PSTATS	53, 55, 56, 65, 66	5
LEGACY	59–62, 70	5
Total		**70**

Tests. ES literals by project: VpHelper 102, OrderFlow 60, Gate0 45, Roll 39, PortMM 23, Dvp 16, Gmm 15, PriceAction 10, fewer elsewhere. These are ES fixtures by design and stay as the bit-identity guard; M-INSTR adds per-root fixtures. Exception: Gate0Tests is edited in PR-M6a (tick, §7).

3.2 Bugs today (fixed independently, PR-M1a–c)

- **F1 (wider than rev 1).** `SymbolRoot` drops digit-led roots ("@6E", "6EZ26" → "") and truncates FX pairs ("EUR/USD" → "EUR", colliding across EUR pairs). `_DvpEngine.RootError` L2445–2447 then treats "" == "" as a match,

so any two digit-led roots pass the same-root check, and roll inheritance (L1840–1842) and the HtfStreams check (L1776–1785) are wrong for them. Fix in PR-M1a (§4.3); `RootError` treats an empty root as an error.

- **F2.** `Category == ESymbolCategory.Future` at L1069, L1866, L1882 misses `FutureRolover`; use the host's `isFuturesCategory` (`_VolumeProfileBase` L1440–1442). PR-M1b.
- **F3 (two bugs).** (a) `_ExitHalfRiskHalfRoll` L107–113 computes `1/PriceScale*BigPointValue`, which is neither a tick nor $/tick; (b) L123 passes `m_firstStopPrice`, a price distance, to `GenerateStopLoss`, which takes a currency amount. Fix: tick = `MinMove/PriceScale`; amount = distance × `BigPointValue` (per contract, matching the strategy's existing per-contract/position mode, VERIFY in the PR). PR-M1c.
- **F4.** `TICK_FALLBACK = 0.01` (L29) with a bare `catch {}` in `resolveTick` (L2287–2301) → Error + throw. PR-M1b.

3.3 MC-enum string uses on main (findings for PR-M1b / allow-list for MI-T14)

file:line	Use	R1 status	Action
`_CustomSignal` L2462	`Category.ToString()` passed to `autoTradingDay` substring tests	violation	PR-M1b: typed `switch` (`ClassOf`)
`_InstrProbe` L67, L74, L75	`Category.ToString()` / reflection	violation	PR-M1e: retire the probe
`_CustomSignal` L279	`TimeZone.ToString()` into a log line	allowed (log)	none
`_CustomSignal` L1883	`res.ToString()` (`EResolution`) into a Warning text	allowed (log)	none
`_Test6` L105	`RollDetection.ToString()` (repo enum `ERollDetection`, `_CustomSignal` L1791) into the A2-63 hash	outside R1 (repo enum) but hash-fragile	PR-M6h: explicit switch, same spellings

4. Detection (typed MC API)

4.1 Members used

Status: **confirmed-PG** = in the MultiCharts .NET Programming Guide (v1.1 §4.2.3 "Properties of the Data Series", pp. 21–22; `mc-sessions-research.md` §6); **compiles** = used in code on main (so the member exists); **VERIFY** = neither, or a behaviour (not the member) is unconfirmed.

Member	Type	
`Info.Category`	`ESymbolCategory`	compi Future Index Money Cash 9 CPCSy Future

		StockC Comm None (box c
`Info.MinMove`, `Info.PriceScale`	double	compi
`Info.BigPointValue`	double	compi curren
`Info.PointValue`	double	compi from I
`Info.Description`	string	compi
`Info.Exchange`	string (data value; parsing allowed)	compi
`Info.Name`	string (data value)	compi
`Info.Resolution.Type` / `.Size`	`EResolution` / int	compi
`Info.TimeZone`	`RequestTimeZone` {Local, Exchange, GMT}	compi
`Info.BVO`	`BuildVolumeOn`	compi

`Info.SessionName`	string	**confir**
		Return
		the ch
		`Equity`
		`(GLOBE`
		(**CONF**
		`01M6HS`
		`01M6HS`

`Info.Request`	`DataRequest`	confir
`Bars.LastBarInSession`	bool	confir **sessio** true a (`01M6H` R14
`Bars.Sessions`	`IROList<SessionObject>` (`.Count`, `[i]`)	**compi** L198– #88 he so the (forun StartC CalcBa (`sameA`
`SessionObject.StartDay/EndDay` (`DayOfWeek`), `.StartTime/.EndTime` (`TimeSpan`)		compi = the zone 17:00– `01M6HS` 18:00– `01M6HS` times

`SessionObject.EndOfSession(DateTime)`, `Sessions.SessionForDate(DateTime)`, an end-of-session bool		forum use th
`Info.ASymbolInfo2`	`MTPA_MCSymbolInfo2` (COM struct)	**confir** “Instr (`MTPA_` `get; }` `/works` struct sectio `Symbol` (for fu L3833 `Symbol` (`MTPA_` **`Symbol`** (`MTPA_` (`DateT` future (doubl L3853 dump `PutOr(` **`Unsupp`** “fields source `Bars.I` symbo (Ettier by Coc table i `Symbol` `Symbol` 1/`Info` 05 vs $ `BigPoi` `Symbol` `Unsupp` `Symbol` contin stocks `MTPA_S` its val `eMTPA_` `Descri` `PutOr(` `Multip` Credit **`MTPA_S`** **`MTPA_M`** **`CONF`** T1 PLI passed failed `_Instr` not ye impor code u

		Instru L973, never value tests/ The M ES and unsup
`Info.Expiration`	`DateTime`	confir compi by ref **neede** Symbol the ro used b
`StrategyCurrencyCode`	`MTPA_MCSymbolCurrency`	compi portfo (_Port _TestF used i (A2). C and **n** ASymbo (CONF INSTR
Per-symbol currency on `Info`	—	no doc guesse preset
Exchange time-zone member, `SessionStartBar`/`SessionEndBar`, “is Default” flag	—	not do
Holidays from code	—	**not av** 2013/2

4.2 Classification (host, exhaustive switch; replaces rev-1 MI-T1)

```
        static EAssetClass ClassOf(ESymbolCategory c) {
          switch (c) {
            case ESymbolCategory.Future: case ESymbolCategory.FutureRolover:
return EAssetClass.Future;
            case ESymbolCategory.Stock: return EAssetClass.Stock;      //
ETF = Stock (rev-3 Q1)
            case ESymbolCategory.Forex: return EAssetClass.SpotFx;
            case ESymbolCategory.Crypto: return EAssetClass.Crypto;
            case ESymbolCategory.Index: return EAssetClass.Index;
            case ESymbolCategory.FutureOption: case
ESymbolCategory.StockOption: case ESymbolCategory.CurrOption:
            case ESymbolCategory.IndexOption: case
ESymbolCategory.MutualFund: case ESymbolCategory.MoneyMarketFund:
            case ESymbolCategory.Cash: case ESymbolCategory.Bond: case
ESymbolCategory.Spread: case ESymbolCategory.CPCSymbol:
            case ESymbolCategory.Composites: case
ESymbolCategory.FutureRoot: case ESymbolCategory.FutureOptionRoot:
            case ESymbolCategory.StockOptionRoot: case
ESymbolCategory.Commodity: case ESymbolCategory.Warrant:
            case ESymbolCategory.None: return EAssetClass.Unsupported;
```

```
            default: throw new InvalidOperationException("M-INSTR: unmapped
ESymbolCategory " + ((int)c).ToString(CultureInfo.InvariantCulture)
              + " (not in spec §4.1); update ClassOf (§4.2) for this MC
build.");
          }
        }
```

All 23 members are listed by name; the `default` throws on a value added in a later MC build (formatting the int is a log, not a parse). It compiles only in PowerLanguage Editor (no test project references PowerLanguage), so it is checked by the PLEditor compile in PR-M2/M3, not by a test. The MC-free core uses only `EAssetClass`.

As merged (rev 2.4, @5c9ccd2). `ClassOf` stays in `_CustomSignal` (CS L2492–2510, region `<instr-host-core>`); `EAssetClass` lives in IP (`<instr-asset-class>`). Two callers:

- `buildProfiles()` calls `InstrumentMcFields.Read(info, ClassOf(info.Category))`. An unmapped `ESymbolCategory` (the `default` throw of `ClassOf`) is **Error + throw `class`** under M4 (A1), same as every other Build refusal. (M3: it was a log-only Warning and a null profile; that path is gone.)
- `classOfData1()` (CS L2708–2723, `TradingDayRoll = Auto`) already writes the Error line (`traceAlways`) and rethrows.

4.3 Root resolution (PR-M1a, fixes F1; rule rewritten in rev 2.2)

1. **Clean:** trim; remove a leading `@`; remove a `#…` continuous suffix; uppercase invariant; remove every `/` ("EUR/USD" → "EURUSD"). In PR-M1a (`_MtfMath.SymbolRoot`, MC-free, no class) the `/` removal applies to every name; **PR-M2 gates it to Spot FX** (`SpotFx`) once the resolver knows the class.
2. **Root:** take the leading digits plus the run of letters that follows them (stop at the first character that is neither). If the letter run has **≥ 2 letters**, its last letter is a month code (`FGHJKMNQUVXZ`) and the character right after the run is a digit, drop that last letter. Nothing else is removed or kept: exchange suffixes (`:XCME`), spaces, year digits and `.`-classes fall away because the run stops there. Examples (MI-T2): `ESZ2024` → `ES`; `'ES 12-24'` → `ES`; `ESZ24:XCME` → `ES`; `ESZ4` → `ES`; `CLZ2026` → `CL`; `MESZ26` → `MES`; `MESH27` → `MES`; `@6E` → `6E`; `6EZ26` → `6E`; `6B` → `6B`; `EUR/USD` → `EURUSD`; `BRK.B` → `BRK`; `Z4` → `Z` (one-letter run: no month drop). (Rev 2.1 steps 3–4, "name minus a trailing month + 1–2 digit year", left `ESZ2024` unchanged; the code on `6853398` (main before #90) L109–119 already used the letter-run form but drops leading digits and does not remove `/`.)
3. **MC's stored root is primary (PR-M2 resolver; rev 2.3, Ettienne; CONFIRMED by Ettienne `_RootDump` 2026-11-03, relayed by Coder).** MC stores the root itself: `Bars.Info.ASymbolInfo2.SymbolRoot` (PG §4.2.3, `/workspace/research/pg11.txt` L901; struct L3828–3867, field L3833). dxFeed **populates it for futures and stocks** (@ES → `ES`, @CL → `CL`, @6E → `6E`, SPY → `SPY`; §4.4.3). The M2 resolver takes it as the **primary** root (trimmed, uppercase invariant); the #90 parser (`_MtfMath.SymbolRoot`, steps 1–2) is a **pure fallback**, used only when `(s.UnsupportedFields & MTPA_SymbolFields.eMTPA_SF_Root) ≠ 0` (typed; also covers `ALL` = -1; `MTPA_SymbolFields` is not `[Flags]` but its values are powers of two, so the bitwise test is valid) **or** `string.IsNullOrWhiteSpace(s.SymbolRoot)`. Never a string or `ToString()` test on the enum (R1). When both roots are non-empty and `!_MtfMath.SameRoot(mcRoot, parsedRoot)`, log one **Warning** `INSTR root-crosscheck mc(<mcRoot>) parsed(<parsedRoot>)`

`name(<Info.Name>)`; **MC's root wins** (it feeds presets, roll inheritance and the same-root checks). The INSTR line records `rootSrc=mc|parsed`. PR-M1a (`_MtfMath`, MC-free, #90) keeps the parser only; it has no access to `Info`. **As merged (rev 2.4):** `InstrumentRoot.Pick` (IP L308–322) implements the choice (MC root trimmed + uppercase; for Spot FX the `/` rule is applied to MC's root too); `Resolve` (IP L298–304) refuses `unknown-field:root` when both are empty. The Warning is logged once at StartCalc (`InstrumentStreams.Start`, IP L934). **Closed 2026-11-04:** the fallback test `(UnsupportedFields & eMTPA_SF_Root) ≠ 0` compiles in PLEditor (MI-T1, §4.1) and the MC runs read `rootSrc=mc` on 6E, ES and AAPL. The parsed-fallback branch itself was not exercised in MC (no symbol flagged the root unsupported); it is covered by the MC-free tests. **MC-internal cross-check (every stream, before presets):** `s.PriceScale × Info.PriceScale = 1` within ±1e-9 relative (ES/CL/SPY 0.01 × 100; 6E 1E-05 × 99999.999999999985 = 0.99999999999999985) and `s.BigPointValue = Info.BigPointValue`. On a mismatch: Error + throw `mismatch:mcPriceScale` / `mismatch:mcBigPointValue` with a clear message, e.g. `M-INSTR: <ENGINE> cannot run on @6E (class Future, root 6E, sessions Default): mismatch:mcPriceScale (ASymbolInfo2.PriceScale 1E-05 × Info.PriceScale 99999.999999999985 = <x>, expected 1 ±1e-9). Check the symbol's settings in QuoteManager.` (§6.3). `MinMove` is equal in all four rows and is logged, not thrown on.

4. **Class-aware handling (PR-M2 resolver only):** stocks keep their share-class suffix (`BRK.B` is its own symbol; the M1a root `BRK` is used only for the MTF/DVP same-root checks), and the `/` gate of step 1 applies to Spot FX only. Presets (PR-M2) match the step-2 root exactly; the rev 2.1 longest-prefix search is dropped because step 2 already isolates the root (`MES` ≠ `ES`, `6E` ≠ `6B`).
5. Other classes when the fallback applies (MC gives no root or flags it): the cleaned name with the same step 2 (crypto `BTCUSD` → `BTCUSD`, `AAPL` → `AAPL`).
6. An empty root (MC and parser both empty) is an error (`RootError` and roll inheritance never compare empty roots).
7. Presets match on root **and** class ("ES" as a stock never gets the ES futures preset).
8. **Cross-check (futures with a preset):** detected tick = `MinMove/PriceScale` and `BigPointValue` equal the preset within 1e-9 relative, else Error + throw `mismatch:tick` / `mismatch:bigPointValue`. The tolerance covers MC's binary priceScale: @6E reports `priceScale(99999.999999999985)` with `minMove(5)` (`01M6HTKV3SV1Q5QDXBFTCH8D3K.txt`), raw tick 5 / 99999.999999999985 = `5.0000000000000009E-05` as .NET Framework `ToString("R")` prints it (17 digits; .NET 8 prints `5.000000000000001E-05`) vs the preset 0.00005, relative error ≈ 1.5e-16 (**CONFIRMED** covered; rev 2.3's `…07e-5` was wrong). **After the cross-check passes the profile carries the preset's exact tick and bigPointValue** (IP L792–793, preset values chosen just above), and `tickValue = tick × bpv` (0.00005 × 125000 = 6.25); without a preset it carries the detected tick and `Info.BigPointValue`. Profile facts on the INSTR line and in the profileHash canon are formatted with `InstrumentText.Fact` = `G15`, invariant (IP L111), so @6E prints `tick(5E-05)` on both runtimes; `InstrumentText.Num` = `"R"` (IP L108) is used **only in diagnostics** (mismatch/refusal texts), whose digits may differ between .NET Framework and .NET 8. The cross-check order and which checks are enforced in M3: §6.2. A detected tick ≤ 0 (or PriceScale ≤ 0) with a preset is `mismatch:tick` (enforced), not `tick`. @ES 25/100 bpv 50, @MES bpv 5, @CL 1/100 bpv 1000, @GC 1/10 bpv 100, @6E bpv 125000 all equal the §8.1 presets.

9. **No preset:** roll rule, contract months, expiry and cost are Unknown; an engine that needs one throws (§6.4 S5).

4.4 VERIFY dump `_InstrDump.Strategy.CS` (PR-M0; typed, no reflection)

Ettienne runs it once on @ES (his live chart, Default sessions), @ES on the "CME Globex" template (the D9a/canary chart), @MES, @CL, @GC, @ZN, @6E, one FX pair, one crypto symbol and one stock, and once in Portfolio Trader. Output: see **§4.4.1** (same directory as the standard MCSignal log, N6 DECIDED). Doubles via `ToString("R", CultureInfo.InvariantCulture)`. Restart MultiCharts after changing any symbol's sessions in QuoteManager (cached until restart, §4.5 setup note).

4.4.1 Output file (N6 DECIDED)

- **Directory** = **the directory of the standard log file the host already creates.** The resolver on `7dbdb1b` is in `_CustomSignal.Strategy.CS`, nested `LogFileSingleton` constructor **L71–91**: `Path.Combine(Environment.GetFolderPath(SpecialFolder.LocalApplicationData), "MCSignalLogs")` (L79), `Directory.CreateDirectory` (L80), a zero-length write to prove access (L82), and on any exception the fallback `Path.GetTempPath()` (L85–88). The resolved file is exposed as `LogFileSingleton.FilePath` (L107) and obtained via `LogFileSingleton.GetInstance` (L93; used at L371 and L395). The same two-step rule is duplicated inline in `TradeJournal` (constructor L168–184, dir L176, fallback L182, `MCJournal_`/`MCEvents_` files).

- **Reuse, no new path.** `_InstrDump` derives from `_CustomSignal` (precedent: `_MtfProbe : _CustomSignal`, `_MtfProbe.Strategy.CS` L89). PR-M0 adds one accessor to `_CustomSignal`, no behaviour change: `protected string LogDirectory() { if (m_logger == null) m_logger = LogFileSingleton.GetInstance(null); return Path.GetDirectoryName(m_logger.FilePath); }` (same lazy pattern as `traceAlways`, L395). It therefore returns the directory of the log file actually created, including the `%TEMP%` fallback. No `const` path, no `C:\Share` path, no new environment lookup in `_InstrDump`.

- **Name**, following the existing `MCSignal<yyyyMMdd-HHmmss>-<pid>.csv` (L76) / `MCJournal_<runId>.csv` / `MCEvents_<runId>.csv` pattern: `MCInstrDump_<symbol>_<yyyyMMdd-HHmmss>-<pid>.txt`, timestamp `DateTime.Now` with `CultureInfo.InvariantCulture`, `<symbol>` = `Bars.Info.Name` with every `Path.GetInvalidFileNameChars()` character and `/` replaced by `_` (`@ES` stays `@ES`; `EUR/USD` → `EUR_USD`). One file per symbol per run (Portfolio Trader: one per symbol).

- **Writing:** one `File.AppendAllText(path, line + Environment.NewLine, new UTF8Encoding(false))` per line (UTF-8 without BOM, as the log, L66), doubles `ToString("R", CultureInfo.InvariantCulture)`. On the first line `_InstrDump` writes one trace line to the standard log, `INSTRDUMP file(<full path>)` (same form as `JOURNAL file(...)`, L2407), and one `Output.WriteLine` with the path. IO errors go to Output and never stop the strategy.

- **Collecting (PowerShell one-liners, no variables).** List the dumps in both possible directories, newest last: `Get-ChildItem -LiteralPath (Join-Path ([Environment]::GetFolderPath('LocalApplicationData')) 'MCSignalLogs'), ([System.IO.Path]::GetTempPath()) -Filter 'MCInstrDump_*.txt' -ErrorAction SilentlyContinue | Sort-Object -

`Property LastWriteTime | Format-Table -Property LastWriteTime, Length, FullName -AutoSize` Make a Desktop folder for them: `New-Item -ItemType Directory -Force -Path (Join-Path ([Environment]::GetFolderPath('Desktop')) 'InstrDump')` Copy every dump into it: `Get-ChildItem -LiteralPath (Join-Path ([Environment]::GetFolderPath('LocalApplicationData')) 'MCSignalLogs'), ([System.IO.Path]::GetTempPath()) -Filter 'MCInstrDump_*.txt' -ErrorAction SilentlyContinue | Copy-Item -Destination (Join-Path ([Environment]::GetFolderPath('Desktop')) 'InstrDump') -Force` Then send the `InstrDump` folder (or attach the files to the PR-M0 thread); they are committed as `tests/InstrumentTests/fixtures/MCInstrDump_*.txt` (data only; nothing in M1 depends on them).

- **A1 (compiles today):** `name, cat (log), catInt, exch, tz (log), minMove, priceScale, bpv, bvo (log), res.Type (log), sessionName`. Description, PointValue and Resolution.Size are already used in compiled code (`_EconomicCalendar` L231/L243/L679, `_CustomSignal` L966/L2245) and are not dumped (`Info.SessionName`, confirmed-PG; record whether it prints "Default" or the template name).

- **A2 Sessions at StartCalc vs first bar:** print `Bars.Sessions` (`n`, then each `StartDay StartTime → EndDay EndTime`) **in StartCalc and again on the first CalcBar**, and the first bar's `Bars.Time[0]` and `Bars.LastBarInSession`. Records: populated at StartCalc (yes/no/null), identical at the first bar, multiple sessions per day kept or collapsed (forum: fixed in MC .NET 9.0). **Result: CONFIRMED** populated and identical in all 13 runs; multiple sessions per day kept (GLOBEX: 9 entries, 17:00–15:15 and 15:30–16:30 each weekday).

- **A3 Session and bar zone:** run @ES and @CL each twice, Format Instrument → Time Zone = **Exchange**, then **Local**. Compare the same bar's `Time[0]` and the printed session times between runs: (i) whether `SessionObject` times follow the chart zone (staff: "main data series" zone); (ii) the exchange offset for NYMEX/COMEX (settles row 4, L2621). **Result: CONFIRMED** (i): session and bar times follow the chart zone (@ES 17–16 Exchange, 18–17 Local); (ii) @CL and @GC show the same 18:00–17:00 either way, see row 4.

- **A4 Multi-series / Portfolio:** with Data2 on a different template, print Data2's sessions (`BarsOfData(2).Sessions`) to confirm per-series sessions; in Portfolio Trader, print each symbol's sessions and `Info.TimeZone` (PT time zone is one global setting). **Result:** per-series sessions **CONFIRMED** (Data1 Default n=5, Data2 GLOBEX n=9, `01M6HSDBFH2D6H21PMYXYKFJHV.txt`); Portfolio Trader: the PR-M0 dump was incomplete (only @ES written, ds2–10 absent, `01M6HSDBFH56AB94QP1SZ4NFP7.txt`); **CLOSED 2026-11-04** by the PT per-symbol test on `5c9ccd2` (below).

- **B (resolved by `_RootDump`, rev 2.3; Ettienne `_RootDump` 2026-11-03, relayed by Coder):**

  - **B1 expiry:** `ASymbolInfo2.SymbolExpiry` is a **placeholder** on continuous futures (2038-01-01 on @ES, @CL, @6E) and stocks (1900-01-01 on SPY). It is used **only when strictly after 1900-01-01 and strictly before 2038-01-01** (i.e. a dated contract); otherwise expiry and roll come from the roll calendar (§8.1) and the roll detector. `Info.Expiration` stays VERIFY and is not needed.
  - **B2 currency:** `ASymbolInfo2.SymbolCurrency` (typed `MTPA_MCSymbolCurrency`): **CONFIRMED** USD for all four; non-USD

throws (Q8). `StrategyCurrencyCode` is not needed.
- **B3 cross-checks:** `SymbolRoot`, `SymbolExchange`, `SymbolCategory`, `MinMove`, `PriceScale`, `BigPointValue` **CONFIRMED** (§4.4.3); `PriceScale` convention: struct = 1/`Info.PriceScale` (the PG "for ES = 1/100" example, L3853, is the struct's); the resolver check is §4.3 step 3. `SymbolCategory` is typed (`MTPA_MCSymbolCategories`) and is logged, not used for classification (§4.2 stays on `Info.Category`).
- `_RootDump.Strategy.CS` is on main (added in `e1f5b11` when `_InstrDump.Strategy.CS` was retired; csproj entry kept). Its four rows are the MI-T2/MI-T3 fixtures (plain strings/ints/doubles in the MC-free core). The original Ettienne dump of 2026-11-03 (relayed by Coder) is the evidence in §4.4.3; the committed probe reprints the same typed fields.

`{i.Category}` etc. are formatted only for the human log; nothing parses them (R1).

4.4.2 PR-M0 dump results (Ettienne's runs, 2026-11-03 17:28–17:43 ET)

Thirteen fixture files from the PR-M0 `_InstrDump` build (rev 2.3: the 6E rerun `01M6HTKV3SV1Q5QDXBFTCH8D3K.txt` replaces the discarded `01M6HSDCCJZXJHNB165KSF6Z5B.txt`; the latter is still listed in `index.txt` but is not committed), indexed in `/workspace/instrdump/index.txt` (to be committed as `tests/InstrumentTests/fixtures/`; the file names are the upload names, not the `MCInstrDump_*` names). Not run yet: @ZN, an FX pair, a crypto symbol (B1/B2 resolved by `_RootDump`, §4.4.3).

File	Run	Facts read from it
`01M6HSDCPGEWFS9VAYZG2C808V.txt`	@ES, Exchange, Default	n=5, Sun 17:00 → Mon 16:0 Thu 17:00 → Fri 16:00; sam StartCalc and first bar; sess end bars 16:00
`01M6HSDBFRQ9ZZFA4NBXZ8QZXC.txt`	@ES, Exchange, GLOBEX template	`sessionName(CME: Equity Futures [E-mini] (GLOBEX)` n=9 (17:00–15:15 + 15:30–1(**session-end bars only at 1 none at 15:15** (R14)
`01M6HSDC526NQSHVTNJG9AQ7A7.txt`	@MES, Exchange, Default	as @ES, bpv 5
`01M6HSDDB20HAY23SAABKQ0MQ5.txt` / `01M6HSDCMNQG1HPR0B9XQWC4S2.txt`	@CL Exchange / Local	18:00–17:00 in both; bars e 17:00 in both; minMove 1, priceScale 100, bpv 1000
`01M6HSDC20TJ572NEKZBKKRBWD.txt`	@GC, Exchange	18:00–17:00, bars end 17:0(minMove 1, priceScale 10, 100
`01M6HTKV3SV1Q5QDXBFTCH8D3K.txt`	@6E, Exchange (rerun 2026-11-03 18:05 ET, after switching @6E to the Exchange	`cat(Future)`, exch CME, priceScale 99999.9999999 minMove 5, bpv 125000; D session Sun–Thu 17:00 → n day 16:00 (Exchange), first 17:06, session-end bars at 1 (`lastBarInSession(True)`): **matches ES**. Chart has `bvo(TickCount)`: a futures on Tick Count must throw R7 (volume basis), so this fi

	& ECN sessions and restarting MC)	is for **instrument and sess** **facts only**, never a volume engine fixture. Replaces `01M6HSDCCJZXJHNB165KSF6Z` (misconfigured custom ses **discarded**; not a fixture)
`01M6HSDBFJ9WRF71G51B82VKCH.txt`	@ES, Local	18:00–17:00; first bar 18:07 session ends 17:00 (= Excha run + 1 h)
`01M6HSDBG7NET9EVPFJ8Z43MR5.txt`, `01M6HSDBWZ22Q03T3AV5GWP4CY.txt`, `01M6HSDBFH2D6H21PMYXYKFJHV.txt`	@ES Data1 Default + @ES Data2 GLOBEX (three identical runs apart from run id)	Data1 n=5, Data2 n=9 (per- sessions); Data1 16:00 bar 2026-10-26 logged as a sess end twice (R14 idempotenc
`01M6HSDBFH56AB94QP1SZ4NFP7.txt`	Portfolio Trader, @ES + @CL, Local	`maxDataStream(10)`, ds2–10 `absent`; only the @ES dump written (superseded: PT pe symbol test passed 2026-11
`01M6HSDC62GCMK5ZANKB61PD5T.txt`	SPY, Exchange	`cat(Stock) catInt(2)`, excl ARCA, `bvo(TickCount)`, Mo 04:15–20:00, session ends 2

CONFIRMED: (1) `Bars.Sessions` is filled at StartCalc and identical on the first bar (all 13); (2) session and bar times follow the chart time zone (@ES 17–16 Exchange vs 18–17 Local; @CL and @GC 18–17 either way); (3) sessions are per data series and multiple sessions per day are kept; (4) @6E priceScale is within the 1e-9 tolerance; (5) SPY is Stock / TickCount / 04:15–20:00; (6) `SessionName` returns the chart's template name. **NEW MC behaviour:** `LastBarInSession` did not fire at the GLOBEX 15:15 break, only at 16:00 (and on the 15:30–16:30 session it fired at 16:00, before the listed 16:30 end, because data stops at 16:00) → R14, MI-T16. **CLOSED 2026-11-04 (Ettienne, `5c9ccd2`): Portfolio Trader per-symbol test.** One PT log (Local tz) for @6E, @ES and AAPL; each stream printed its own root, tick, bpv, sessions and calendar:

- **ES:** tick 0.25, bpv 50, `cost(17.00 src=Ettienne 2026-10-31)`, `floor(14)`, `rth(09:30-16:15 src=EquityIndexLegacy)`, profileHash `e1927ccf1ea6f41e` (Local tz; reproduced on the box from the Default list `Sun-Thu 18:00→17:00`).
- **AAPL:** `class(Stock)`, `preset(none)`, tick 0.01, `calendar(Nyse)`, `rth(09:30-16:00 src=ChartDaySession)`, profileHash `41886799d411dd0d`.
- **@6E:** profileHash `9523b25025f9ed5b` (Local, sessions 18:00→17:00 ET) vs `55576df6a8a99c05` on the Exchange-tz chart: the evidence for D2 (§12.4).

The PR-M0 question (did @CL not start?) is moot. Ettienne's earlier @6E rerun is closed (below).

CLOSED (rev 2.3): @6E zone. Ettienne's QuoteManager had @6E on **Use Custom Sessions**, Sun–Thu 18:00 → next day 17:00, with Session TimeZone = **Exchange (CT)**: ET hours entered as CT. A user configuration error, not MC behaviour (CME 6E trades 17:00–16:00 CT like ES, research `/workspace/m-instr/mc-session-templates-research.md` "Observation: 6E", https://www.cmegroup.com/trading/why-futures/welcome-to-cme-fx-futures.html). He switched @6E to the Exchange & ECN sessions and restarted MC; the rerun `01M6HTKV3SV1Q5QDXBFTCH8D3K.txt` reads Sun–Thu 17:00 → 16:00 Exchange, first bar 17:06, session ends 16:00, matching ES. It is the 6E fixture; `01M6HSDCCJZXJHNB165KSF6Z5B.txt` was misconfigured and is discarded (dropped from the fixture list).

4.4.3 `_RootDump` results (Ettienne `_RootDump` 2026-11-03, relayed by Coder; dxFeed)

Symbol	root	exch	cat	ccy	SymbolExpiry	m
@ES	ES	CME	FUTURE	USD	2038-01-01	25
@CL	CL	NYMEX	FUTURE	USD	2038-01-01	1
@6E	6E	CME	FUTURE	USD	2038-01-01	5
SPY	SPY	ARCA	eMTPA_MCSC_STOCK	USD	1900-01-01	1

`SymbolName` = `Info.Name` in every case. `MTPA_SymbolFields` values as in §4.1.

4.5 Sessions from the chart (R12; DECIDED Q1/Q2)

What MC does (research doc `/workspace/m-instr/mc-sessions-research.md`). Each data series has its own session setting (Format Instrument → Settings → Sessions = **Default**, i.e. the symbol's QuoteManager sessions, or a named template); in Portfolio Trader each symbol has its own session but the time zone is one global setting. Scripts cannot set sessions. Data outside sessions is filtered out; time bars always break on session boundaries and align to the session start; `Time` is the bar close. Holidays are removed by QuoteManager holiday lists and **cannot be read from code**. **Templates (research `/workspace/m-instr/mc-session-templates-research.md` §1, §3, §7):** a chart uses Default (the symbol's QuoteManager sessions: exchange sessions, custom sessions, a custom template or the symbol dictionary) or a named template; templates, built-in ones included, are editable in **QuoteManager → Tools → Session Templates** (Add/Edit/Delete, rows with Session Time and Session End; https://www.multicharts.com/trading-software/index.php?title=Setting_Custom_Session_Templates, https://www.multicharts.com/discussion/viewtopic.php?t=5890), but not from a script (staff, https://www.multicharts.com/discussion/viewtopic.php?t=53293). **"Combined" templates mean two sessions per day** (overnight + day, two daily bars; https://www.multicharts.com/discussion/viewtopic.php?t=45183), not one merged 23-hour session; "Open Outcry"/"Open Auction" templates describe pit sessions closed in 2015 and survive only as RTH-style windows.

Setup note (MC caches sessions). MultiCharts caches QuoteManager symbol sessions until a restart: after changing a symbol's sessions in QuoteManager, **restart MultiCharts** before running charts, dumps or tests (the @6E rerun, §4.4.2, needed it).

Environment caveat (not an engine rule). The built-in **"CME: Equity Index Futures [E-mini] (GLOBEX)"** template is stale: its 15:15–15:30 CT gap is the halt CME removed on 27/28 June 2021 (https://www.cmegroup.com/notices/electronic-trading/2021/06/20210621.html), and its 15:30–16:30 row matches the hours **before November 2012** (2012–2021 sessions ended 16:15, https://www.cmegroup.com/tools-information/lookups/advisories/market-data/20121112.html; pre-2012 sheet https://www.dormantrading.com/TraderTools/CMEeminisp500.pdf); current ES hours are 17:00–16:00 CT (https://www.cmegroup.com/markets/equities/sp/e-mini-sandp500.html). This template is on the **D9a/canary chart** (§4.4), so that chart has a session boundary at 15:15 and a listed end at 16:30. Engines **take sessions as given** (R12) and do not correct or warn about it; changing the canary chart's template is Ettienne's environment decision and would change MI-T15 baselines.

Reading. `SessionReader` (host) reads `BarsOfData(n).Sessions` into the MC-free struct `_MtfMath.Session` (#88 head, L47–52: `StartDay, StartTime, EndDay, EndTime`) **in StartCalc** (populated there, CONFIRMED §4.4.2) and **re-checks on the first CalcBar of each stream** (must be identical, else Error + throw `sessions:changed`, S1). Session ends follow R14 (from the list's `EndTimes`, not `LastBarInSession` alone). Null or empty sessions → throw `sessions:none` for every session-using engine (§6.4 S1). Template name: `Info.SessionName` on the INSTR line, log only. Whether the chart uses Default is not exposed (not documented) and is not checked: the chart's template is the truth.

Zone. Session times are in the bar time zone of the stream, i.e. the chart's Format Instrument time zone (**CONFIRMED** A3, §4.4.2: @ES 17:00–16:00 on Exchange, 18:00–17:00 on Local, bar times shift by the same hour; @CL/@GC identical either way). No conversion is needed before deriving. ET-anchored logic that remains (EVENT rows, the frozen G0/REGGATE trial windows) converts bar time to ET as today: Local → `TimeZoneInfo.Local`, GMT → UTC, Exchange → the `exchangeTimeZoneId` map (L2614–2626, kept only for this; its NYMEX/COMEX entry stays Eastern: A3 shows MC's exchange time for @CL/@GC equal to ET on Ettienne's setup, row 4).

Derived fields (MC-free, `<instr-core>`): - `tradingDayKey(t)` = the calendar date, in session time, of the **end** of the session that contains bar close `t` (start exclusive, end inclusive). Several sessions ending on the same date share the key (e.g. the GLOBEX template 17:00–15:15 + 15:30–16:30, `01M6HSDBFRQ9ZZFA4NBXZ8QZXC.txt`). On the CME Globex template (17:00–16:00 CT) this equals today's `GlobexTradingDayEt` (18:00 ET roll) for every bar; on a 24×5 FX template ending 17:00 ET it equals today's ET+7h rule; on a 00:00 UTC crypto template it equals the UTC date. A bar outside every session (should not occur, MC filters) → Error + throw `sessions:bar-outside`. - `dayOpen` = start of the first session of the trading day (TOD minute-of-session; replaces 18:00 ET at VPFUT L489–493 and the default at L949). - `sessionEnd` = end of the last session of the trading day, from the list (R14; on the GLOBEX template 16:30 even though the last bar closes at 16:00) (FLOW tick-chunk boundary, replaces `TickChunkBoundaryEt` 17:00 and the rev-1 `rollTime - 1h`). - `hasOvernight` = some session of the trading day crosses midnight in session time (`StartDay ≠ EndDay`). Globex 17:00 CT Sun →

16:00 CT Mon: true; an 08:30–15:15 day session: false. For 24-hour single-day templates (e.g. 00:00–24:00 crypto) the value is fixed in PR-M4 from the A2 output (**VERIFY**). - `rthSplit` = the RTH segment inside the trading day: **when the chart's sessions are day-only (`!hasOvernight`), the chart session is the RTH**. For a 24-hour chart (N1 DECIDED): ES/MES/NQ/MNQ use today's equity-index split from `_CmeSessionClock` (RTH 09:30, value end 16:00, session flag 16:15 ET), logged `rth src=EquityIndexLegacy`; any other root on a 24-hour chart has `rthSplit = null` (S7). No RTH window in any preset. - `sessionsPerWeek`/`sessionsPerYear`: annualisation stays a class constant (252 futures/stocks, 365 crypto, 260 FX **VERIFY**), not derived.

Multi-stream. Each stream has its own derived fields. The #86/#88 DV/MTF Warning when Data2..N sessions differ from Data1 (`checkHtfSessions`, #88 head L1802–1822) stays unchanged, including its MC-1439 / Realtime-History Matching caveat: with RHM on and different session ends, the strategy stalls after the shorter session's end until the auxiliary series prints. In Portfolio Trader, mismatched bar-close times across symbols can make the portfolio money-management signal run several times per bar (MC staff); M-PORT behaviour is unchanged, and the INSTR line shows each symbol's sessions so a mismatch is visible.

Removed because they existed only for hardcoded times (PR-M4 unless noted): `_CustomSignal` L1888–1889 (`"Eastern Standard Time"` + `GlobexTradingDayEt` for the roll key → `tradingDayKey`); `autoTradingDay` L2581–2587 + L2462 (→ `tradingDayKey`); VPFUT `GlobexDate` key L361 and TOD 18:00 L489–493; `_CmeSessionClock` Globex open/close/halt defaults L48–61 as *trading-day* inputs; `TickChunkBoundaryEt` (PR-M6b). The exchange-zone map stays only for ET conversions (above). Holiday calendars stay (they are not session times; §6.4 S4, PR-M8).

ES bit-identity. On the CME Globex template (D9a and canary charts, `_DvpEngineTest` L28–48, `_TestGate0` L42) every derived day key, day open and session end equals today's constant. On a chart with another template (e.g. Ettienne's live @ES Default sessions) results follow that chart; this is the intended behaviour of the decision and is logged on INSTR, not warned.

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5. Data model

5.1 `InstrumentProfile` (immutable; MC-free core `<instr-core>` in `_InstrumentProfile.Function.CS`)

Field	Type	Source
`assetClass`	`EAssetClass { Future, Stock, SpotFx, Crypto, Index, Unsupported }`	detect (§4.2)
`root`	string	`ASymbolInfo2.SymbolRoot` (CONFIRMED), parser only as fallback (§4.3 step 3)
`presetId`	string or null	preset
`tick`	double	preset tick when a preset matched (after the §4.3 cross-check), else `MinMove/Info.PriceScale`

`bigPointValue`	double	preset bpv when a preset matched else `Info.BigPointValue`
`tickValue`	double	tick × bigPointValue
`currency`	ISO code	`ASymbolInfo2.SymbolCurrency` (CONFIRMED, §4.4 B2)
`sessions`	`_MtfMath.Session[]`	`Bars.Sessions` (§4.5) via `SessionReader` (IP L996–1006)
`sessionName`	string	`Info.SessionName`
`barZone`	`RequestTimeZone` + resolved tz id	`Info.TimeZone` (+ map for Exchange)
`tradingDay`	derived from `sessions`	§4.5
`dayOpen`, `sessionEnd`, `hasOvernight`	derived	§4.5
`rthSplit`	(start, end) or null, + `rthSource ∈ {ChartDaySession, EquityIndexLegacy, None}`	§4.5, N1
`calendar`	`ECalendar { CmeEquityIndex, Nyse, NyseStandIn, Observed, None }` (IP L33; + CME product groups when verified, PR-M8)	preset; no preset: Stock → `Nyse`, Unsupported → `None`, else `Observed` (IP L252)
`contractMonths`	month mask	preset
`expiry`	`EExpiryRule` + params	preset; a dated contract's `SymbolExpiry` when strictly inside (1900-01-01, 2038-01-01) (§4.4 B1)
`roll`	`ERollRule` + params	preset
`sessionsPerYear`	int	class
`cost`	`CostModel { roundTripCents (long), source }`	`InstrumentConfig.costs`
`costFloorTicks`	int	§9 (integer cents); computed **only when a preset matched**, with the preset tick/bpv (IP L778–779)
`cluster`	string	preset
`volumeBasis`, `volumeQuality`		`Info.BVO` + class
`dataNotes`	string or null	`InstrumentConfig.dataNotes[root`

`sizingUnit`	Contracts / Shares / Units	class

5.2 Presets: exchange facts only (no times)

`InstrumentPresets`: static table in `<instr-core>`, one row per root (§8): tick, bigPointValue, currency, contractMonths, expiry, roll, calendar, cluster, source. Adding a root = a one-row PR with sources.

5.3 `InstrumentConfig` (protected on `_CustomSignal`; no inputs)

- `enabled` (when false: `buildProfiles()` skipped — no profiles, no refusals, no INSTR / profileHash; host tick from MinMove/PriceScale).
- `overrides`: typed `InstrumentPresetRow`, null by default (fix an exchange fact without a code PR; contains no session fields).
- `costs`: **asset class** + **root** → round trip in **cents** (`long`) (**D1 DECIDED**, Ettienne 2026-11-04, §12.4). One `costSource` string for all (default `Ettienne 2026-10-31`, printed `cost(17.00 src=…)`); values > 0.
  - **Key format:** `<Class>:<ROOT>`, where `<Class>` is the `EAssetClass` name exactly as `InstrumentText.ClassName` prints it (`Future`, `Stock`, `SpotFx`, `Crypto`, `Index`; `Unsupported` is not allowed) and `<ROOT>` is the resolved root, trimmed uppercase, as on the INSTR `root(…)` field. Code builds keys with one helper, `InstrumentConfig.CostKey(EAssetClass, string root)`, so no code parses a class name back from text; `Validate` accepts a key only when it equals `CostKey(c, r)` for some class `c ≠ Unsupported` and a non-empty uppercase `r`. A root-only key (`ES`) or a misspelt class is invalid → `unknown-field:config` (no silent migration of old keys).
  - **Defaults map:** ES 1700 → `Future:ES` 1700; MES 250 → `Future:MES` 250. `costSource` unchanged.
  - **Lookup:** `CostFor(assetClass, root)` reads only `CostKey(profile class, profile root)`.
  - **Class mismatch:** when no row exists for the symbol's own class but a row exists for the same root under another class (e.g. a stock `ES` while `Future:ES` is configured), that row is **not used**: the profile has no cost (`cost(Unknown)`), the floor is Unknown with its reason (`floor(Unknown src=cost-class-mismatch)`, §6.1/§9), and one Warning per stream at StartCalc names it: `INSTR ds(<n>) cost row Future:ES not used: symbol class Stock root ES`. Never silently applied, never silently dropped.
  - **Canon:** `InstrumentConfig.Canon()` keeps its form with the new keys, sorted ordinal: `costs=[Future:ES:1700,Future:MES:250,]` (cents after the last `:`). The profile Canon is unchanged (`cost=<cents>` or `-`; floor `-1` when Unknown; the src reason is printed, not hashed), so the pinned ES/MES/6E profileHashes do not change; only a profile that today shows a cross-class cost (stock `ES`: `cost=1700` → `cost=-`) gets a new hash.
  - **configHash:** not affected. On `5c9ccd2` `InstrumentConfig.Canon()` (IP L413–424) is called only by tests (IT L457, L546); it is not part of the RUN line (CS L2296–2302: `inputs[…]`, `event[EventConfig.Canon()]`, profileHash suffix) or of any engine `configHash` (e.g. `_DvpEngine` hashes `DvpConfig.Canon()` only). If a later PR (PR-M7 cost floor) puts `InstrumentConfig.Canon()` into a hash, it must be with the class+root keys from the start.
  - **M3 behaviour (as merged, superseded by PR-M4):** keyed by **root**

only (IP L398, `CostFor(root)` L426–430; keys trimmed uppercase): a stock and a future with the same root share a cost (a stock ticker `ES` shows `cost(17.00 …)`, IT L609–611). M3 limits the damage by computing the floor only for a matched preset (§9).

- `applyCostFloor` (PR-M7, default false).
- `dataNotes`: root → note string.
- `enabled = false` skips `buildProfiles()` with one Info line `INSTR off (InstrumentConfig.enabled = false): no profiles, no cross-checks` (CS L2392–2396 @5c9ccd2): no INSTR lines, no RUN suffix.
- An invalid config (`Validate() ≠ null`) is the log-only refusal `unknown-field:config`; its `overrides` row is not trusted for the cross-check (§6.2).
- `Validate()`, `Canon()` as in existing configs; `profileHash` outside every `configHash` (rev-3 Q7).

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6. Logging, application and refusal

6.1 Lines (`engineSink(level)`, one `File.AppendAllText` per line, InvariantCulture)

- `INSTR ds(<n>) name(<Info.Name>) cat(<Info.Category label>) class(<EAssetClass>) root(<r> rootSrc=mc|parsed) preset(<id>|none) exch(<Info.Exchange>) tz(<setting>→<zone id>|unresolved|unavailable(<ExceptionType>)) sessionName(<Info.SessionName>) sessions(n=<k> <Day hh:mm>→<Day hh:mm>; …) [xsessions(<exchange zone id> n=<k> …; rth <hh:mm-hh:mm>|none) — PR-M4, D2] dayKey(SessionEnd) dayOpen(hh:mm) sessionEnd(hh:mm) overnight(yes|no) rth(<start>-<end> src=EquityIndexLegacy|ChartDaySession) | rth(none src=None) tick(<G15>) bpv(<G15>) tickValue(<G15>) ccy(USD) bvo(<label>) calendar(<ECalendar>[; <note>]) months(<letters>) expiry(<rule>|none) roll(<rule>[(n)]|none) cost(<c.cc> src=<costSource>)|cost(Unknown) floor(<n>)|floor(Unknown) cluster(<c>|none)[dataNotes(<note>)] profileHash(<16 hex>)` at **Info**, once per stream after the first-bar session read (`InstrumentProfile.InstrLine`, IP L846–872 @5c9ccd2). `dataNotes(…)` appears only when `InstrumentConfig.dataNotes` has the root; it sits after `cluster` and before `profileHash`. `cat`, `bvo` and the tz setting are MC enum labels formatted by the host for the log only (R1; `unavailable(<ExceptionType>)` when the typed read throws, CS L2449–2474). The tz zone part is the resolved `TimeZoneInfo.Id`, `unresolved` when no zone resolves, or `unavailable(<ExceptionType>)` (CS `instrZoneLabel` L2465–2471). **Pinned (rev 2.4; the tests pin these exact strings).** ES, CME GLOBEX 9-entry list (IT T12 L537–539): `INSTR ds(1) name(@ES) cat(Future) class(Future) root(ES rootSrc=mc) preset(ES) exch(CME) tz(Exchange→Central Standard Time) sessionName(test) sessions(n=9 Sun 17:00→Mon 15:15; Mon 15:30→Mon 16:30; Mon 17:00→Tue 15:15; Tue 15:30→Tue 16:30; Tue 17:00→Wed 15:15; Wed 15:30→Wed 16:30; Wed 17:00→Thu 15:15; Thu 15:30→Thu 16:30; Thu 17:00→Fri 15:15) dayKey(SessionEnd) dayOpen(17:00) sessionEnd(16:30) overnight(yes) rth(09:30-16:15 src=EquityIndexLegacy) tick(0.25) bpv(50) tickValue(12.5) ccy(USD) bvo(Volume) calendar(CmeEquityIndex) months(HMUZ) expiry(ThirdFridaySoq) roll(EquityIndexCustomary) cost(17.00 src=Ettienne 2026-10-31) floor(14) cluster(EQIDX) profileHash(020deca78d6392b7)` 6E, same template, NyseStandIn calendar (IT T13 L475; tz label as the test passes it): `INSTR ds(1)`

`name(@6E) cat(Future) class(Future) root(6E rootSrc=mc) preset(6E) exch(CME) tz(Exchange) sessionName(test) sessions(n=9 Sun 17:00→Mon 15:15; …same 9 entries as ES…) dayKey(SessionEnd) dayOpen(17:00) sessionEnd(16:30) overnight(yes) rth(none src=None) tick(5E-05) bpv(125000) tickValue(6.25) ccy(USD) bvo(Volume) calendar(NyseStandIn; calendar: NYSE stand-in (CME calendar pending M8)) months(HMUZ) expiry(ThirdWedMinus2Bd) roll(BusinessDaysBeforeExpiry(5)) cost(Unknown) floor(Unknown) cluster(FX_EUR) profileHash(8e8b3e9b0e230e12)` The 6E hash `8e8b3e9b0e230e12` is for the **synthetic** 9-session GLOBEX list the test passes. **MC result (MI-T15, Ettienne 2026-11-04):** the real @6E chart (Exchange tz, Default session, BVO TickCount) printed `tick(5E-05) bpv(125000) tickValue(6.25)`, the NyseStandIn calendar note, `roll(BusinessDaysBeforeExpiry(5))` and profileHash **`55576df6a8a99c05`**; the box (.NET 8) reproduces it byte-for-byte from the canon with `sessions=n=5 Sun 17:00→Mon 16:00; … Thu 17:00→Fri 16:00`, so the hash is **runtime-stable** (.NET Framework = .NET 8). It differs from the pin only by the session list.

- **Number formats.** Profile facts on the INSTR line and in the hash canon use `InstrumentText.Fact` = `ToString("G15", InvariantCulture)` (IP L111): `5E-05`, `0.25`, `12.5`. `InstrumentText.Num` = `"R"` (IP L108) is used only inside diagnostics (mismatch and refusal texts), e.g. the raw 6E tick `5.0000000000000009E-05` (.NET Framework; .NET 8 prints `5.000000000000001E-05`). Diagnostic digits may differ between runtimes; profile text and hash may not.
- **profileHash** = FNV-1a 64 over the canonical profile text (`Canon`, IP L825–835; `ProfileHash` L836–841): class, root, preset, tick, bpv, ccy, calendar, months, expiry, roll, cost cents, floor (-1 when Unknown), cluster, sessionsPerYear, the session list and the RTH split (**in exchange time**, D2, as merged in PR-M4). Not hashed: name, exchange, sessionName, the tz/cat/bvo labels, costSource, `dataNotes`, tickValue and the derived day fields. **RUN suffix (M4):** the RUN line ends with `profileHash(<ds⩾<hash>;…)` over the **started** streams only (a stream counts as started only once `Build` returned a profile; every refusal throws, so there is no `none` entry any more); empty when M-INSTR is off or no stream started (`InstrumentStreams.RunSuffix`, IP L953–963; appended at CS L2302 after the `configHash` part). It is **outside every `configHash`** (rev-3 Q7); IT T12 L544 pins the format.
- **Log-only line (M3 only; removed in M4):** at **Warning**, `INSTR ds(<n>) log-only, not enforced until PR-M4: <refusal>`. Under M4 every refusal is Error + throw; the log-only helper and the null-profile path are gone. Failed typed / session reads also Error + throw (`instrReadFailed`, CS).
- **Root cross-check** (§4.3 step 3), once per stream at StartCalc, **Warning**: `INSTR root-crosscheck mc(<r>) parsed(<p>) name(<n>)`.
- **Unexpected exception inside a guarded step**: `INSTR ds(<n>) unexpected <ExceptionType>: <message>` at **Error**, then rethrown (IP L925).
- **M-INSTR off**: `INSTR off (InstrumentConfig.enabled = false): no profiles, no cross-checks` at Info (CS L2394); no INSTR lines, no RUN suffix.
- **D2 (DECIDED, Ettienne 2026-11-04; PR-M4): profileHash over sessions in exchange time.** Today `Canon` hashes `InstrumentText.Sessions(sessions)` in the **bar** time zone, so one instrument hashes differently per chart/PT zone: @6E Exchange-tz chart (Sun 17:00→Mon 16:00 … Central) `55576df6a8a99c05`, @6E in Portfolio Trader on Local (18:00→17:00 ET) `9523b25025f9ed5b`. From PR-M4:

  - **Zones (as merged, PR-M4 @d3c0c1a).** Bar zone = `strictBarTimeZone(Bars.Info)` (CS L2954+): Local → Local, GMT → Utc, Exchange → the strict exchange lookup, anything else → null (no Local fallback). Exchange zone = `exchangeTimeZoneId(Info.Exchange)` (virtual) → static `exchangeZoneIdOf` (CS L2587–2615, `<instr-host-core>`): split the name on non-alphanumerics, exact case-insensitive token match, **first matching token wins**. Chicago → `Central Standard Time`: CME, XCME, GLOBEX, CBOT, ECBOT, CBT, XCBT, CBOE. New York → `Eastern Standard Time`: NYMEX, COMEX, NYSE, NASDAQ, ARCA. Looked up **whatever the chart's `Info.TimeZone` setting is**. The old `findTz` Local/`America/New_York` fallbacks are not on the profile path.
  - **Refusal, never a silent fallback.** Bar zone unresolved → `tz:bar`; `exchangeZoneIdOf` returns null or the id does not resolve → `tz:exchange` (§6.3; Build refuses and throws — there is no `profileHash(…=none)` path under M4). Action text: "add the exchange to exchangeZoneIdOf" / "set the chart time zone". A symbol on an exchange outside the map (FX, crypto feeds, and composite names such as NYSEARCA / NASDAQGS / CMEGLOBEX / AMEX / BATS / CFE) is refused until its exchange is mapped (**A3 OPEN**, §12.4).
  - **Conversion.** Each session's start and end (`DayOfWeek` + time) is converted bar zone → exchange zone on the trading week of the first bar (Sunday on or before the first bar's date, bar time), via `TimeZoneInfo.ConvertTime`; identity when the two zones have the same id. Same-DST zone pairs (ET↔CT) give the same text in any week. **DST coverage in IT (M4):** summer week, spring-forward gap, and fall-back ambiguous hour (read as standard) all keep 6E Default pin `55576df6a8a99c05`. A **GMT** chart remains **VERIFY** (does MC shift GMT sessions across DST?); one MC GMT run is still open (§12.4 A3).
  - **What is hashed.** `sessions=` in exchange time, and `rth=` recomputed by `RthSplit.Of` on the exchange-time list (rth **is** in `Canon`, IP L834; a `ChartDaySession` rth is session-derived; `EquityIndexLegacy` is fixed ET constants and is unchanged). The zone id itself is **not** hashed (Windows `Central Standard Time` vs `America/Chicago` on .NET 8 would break runtime stability). `dayOpen`/`sessionEnd` are not hashed and stay in bar time.
  - **What engines and the INSTR line use.** The profile keeps `sessions`, `rthSplit`, `dayOpen`, `sessionEnd` in **bar time** for the engines (R12, bar times are in bar zone). The INSTR line keeps `sessions(…)`, `dayOpen`, `sessionEnd`, `rth(…)` in bar time and adds `xsessions(<exchange zone id> n=<k> …; rth <hh:mm-hh:mm>|none)` = exactly the hashed exchange-time text.
  - **Pins as merged (box + IT; confirm equal hashes in MC on Exchange chart vs PT Local).** @6E Default `55576df6a8a99c05` on **any** chart/PT zone (PT Local `9523b25025f9ed5b` goes away); @ES Default (Sun–Thu 17:00→16:00 CT, EquityIndexLegacy rth) `6cff053d1b215324` on any zone (PT Local `e1927ccf1ea6f41e` goes away); AAPL `41886799d411dd0d` unchanged when the PT PC is on ET (bar zone = exchange zone). The IT pins ES `020deca78d6392b7` / 6E `8e8b3e9b0e230e12` stay **only if** the test declares bar zone = exchange zone; every test pin built from a non-exchange bar zone changes, and the tests gain the invariance checks of §11.1 item 9.
- **Floor-Unknown reason (DONE in PR-M4):** `floor(Unknown src=no-preset|no-cost|cost-class-mismatch|config-invalid)` (§9, `FloorSource`).
- `CFG <ENGINE> applied{…} undefined{output=reason;…} degraded{mode}` at Info, once per engine.

- `SKIP <ENGINE> reason(<code>) …` at **Info**.
- Mismatches (and every other enforced refusal, §6.2) go out at **Error** before the throw.

6.2 Application

`buildProfiles()` runs in `StartCalc` for instrument fields; session-derived fields are filled on the first CalcBar of each stream, and engines whose configuration needs them (TOD, AVWAP session date, FLOW chunk plan, RANGE/WEEKLY session sources, VPFUT day key) finish configuring there, before their first update. Engines get the profile through their existing config objects; engines never read `Bars.Info` or `Bars.Sessions` directly.

As merged in PR-M4 (#98, @d3c0c1a); every refusal enforced globally (A1). Engines still do not read a profile yet (they move on from PR-M5).

- `buildProfiles()` (CS L2406+) runs in StartCalc **before** the engines start. Per stream: typed read (`InstrumentMcFields.Read`), `InstrumentProfile.Build` via `InstrumentStreams.Start` / `Guard`, StartCalc sessions via `SessionReader`. `profilesFirstBar()` runs on the first CalcBar, before the RUN line: re-reads sessions, `InstrumentSessions.Require` (S1), derives day key/open/end/RTH split (`WithSessions`) and prints the INSTR line.
- **`Build` is two-phase** (IP L858–922 @d3c0c1a). **Phase 1, the cross-checks**, runs only when the class is not Unsupported: root pick → `Checks.McInternal` (`mismatch:mcPriceScale` / `mcBigPointValue`) → raw tick → preset choice (valid `overrides` row matching class + root, else `InstrumentPresets.Find`) → `Checks.Preset` when a preset matched (`mismatch:tick` / `mismatch:bigPointValue`) → the cost floor when the config is valid and a preset matched (`mismatch:tickValue`, §9). **Phase 2, the refusals** (all enforced): `class` → `unknown-field:config` (invalid `InstrumentConfig`, incl. old root-only cost keys) → `unknown-field:root` → `tick` (PriceScale ≤ 0 / tick ≤ 0, no preset) → `currency` → `calendar` (when the profile has an expiry or roll rule, R6) → `tz:exchange` / `tz:bar`.
- **Enforcement (M4, A1).** `InstrumentStreams.Guard` buffers the Error line; **every** `InstrumentException` writes it and rethrows (the study stops). There is no log-only path and no null-profile / `profileHash(…=none)` path. Any other exception writes `INSTR ds(n) unexpected …` at Error and rethrows. **`Start` sets `m_started` / `m_pending` only after `Guard(Build)` returns** — a refused stream is never marked started (`InstrumentStreams.Start`, IP).
- **Session refusals on the first bar** (`sessions:none`, `sessions:changed`, `sessions:bar-outside`, …) are likewise Error + throw under M4 (`FirstBar` → `Guard(WithSessions)`).
- **What is refused vs what still runs (confirmed @d3c0c1a).** Refused (Error + throw): Unsupported / unmapped class; non-USD; unresolved bar or exchange zone (incl. unmapped exchange → `tz:exchange`); PriceScale ≤ 0 / tick ≤ 0; missing or changed sessions; failed typed or session reads; invalid `InstrumentConfig` (incl. old root-only cost keys → `unknown-field:config`); `mismatch:*`; `calendar` / `months` when rules need them. **Still runs:** futures with no preset (cost / floor Unknown, `src=no-preset` or `no-cost`); USD stocks on NYSE / NASDAQ / ARCA (mapped → Eastern).

6.3 Exception format

`M-INSTR: <ENGINE> cannot run on <name> (class <c>, root <r>, sessions <sessionName>): <reason>. <action>.` Reason codes (`InstrumentReason`, IP L46–56 @5c9ccd2): `class`, `unknown-field:<f>` (`root`, `config`, `cost`), `tick`, `mismatch:<f>` (`tick`, `bigPointValue`, `mcPriceScale`, `mcBigPointValue`, `tickValue`), `volume-basis`, `currency`, `sessions:none`, `sessions:changed`, `sessions:bar-outside`, `sessions:no-rth-split`, `sessions:window-not-covered`, `calendar`, **`months`** (rev 2.4: `ContractFor` finds no listed contract month within 24 months, action "check the preset's contract months", IP `ContractFor` L519–530; tested IT T4 L241–242), and from PR-M4 (D2) **`tz:bar`** (the bar time zone cannot be resolved) and **`tz:exchange`** (no exchange zone for `Info.Exchange`, or the zone id does not resolve). Under M4 every reason above is enforced (§6.2, A1).

6.4 R13 applied: every throw, degraded mode and undefined output

Throw rules | # | Condition | Engines | Action text | |—|—|—|—| | S1 | `Bars.Sessions` null/empty at StartCalc or on the first bar; or StartCalc and first-bar reads differ; or a bar outside every session | every session-using engine (TOD, AVWAP, VPFUT, FLOW, RANGE, WEEKLY, DAILY, DVP, G0, REGGATE, ROLL keying) | "check the chart's session template" | | S2 | An engine whose purpose needs an overnight segment, on a chart with `!hasOvernight` | none on main (every overnight-specific item is an *output*, see U1–U2) | — | | S3 | A frozen pre-registered window (G0 09:30–16:00 ET, `_Gate0` L474/L614/L1037…; REGGATE 09:30–16:00 ET, `_RegimeGate` L45–48) is not inside the chart's sessions (converted to ET) | G0, REGGATE | "use a session template that covers 09:30–16:00 ET" | | S4 | An engine needs a holiday or early close **in advance** and the root has no verified calendar (rev 2.4: `NyseStandIn` for CL/GC/ZN/6E is a stand-in, not verified; it serves the expiry/roll business-day rules only. A preset with roll/expiry rules and no calendar is refused `calendar` at Build, R6; enforced since M4) | REGGATE (early-close exit), G0 (early flat), VPFUT `HaltEt`/`RthCloseEt`/`IsGlobexClosedDay` hooks (L1635–1671), `_CmeSessionClock` holiday use | "add a verified calendar for (PR-M8)" | | S5 | A required preset field is Unknown (`roll`, `contractMonths`, `expiry`, `cost`) | ROLL Table/SeriesWindow, explicit EVENT own-expiry queries (not Step/Classify; 2026-11-06), DVP roll table, COST/cost floor | "add a preset row / cost for " | | S6 | Class Unsupported in the rev-3 matrix | per matrix | rev-3 reason text | | S7 | An engine needs `rthSplit` and it is null (24-hour chart, root not ES/MES/NQ/MNQ; N1) | VPFUT RTH profile/value window/IB, AVWAP `rth`, WEEKLY `RthSessions`, DVP RTH features | "use a day-session template for " | | S8 | tick ≤ 0; detected ≠ preset; futures on Tick Count (R7); non-USD (Q8) | all | — |

Approved degraded modes (only these; each from a DECIDED answer): D1 VP on FX/CFD/crypto as visual/level only, nothing feeds Holm (rev-3 Q5); D2 G0 on Index price-only parts (rev-3 matrix note 6; G0 is ES-canary only in practice); D3 EVENT on USD FX pairs, US events as a filter (rev-3 Q11); D4 FLOW on crypto, single venue, Warning (rev-3 reason 12); D5 PriceAction on Bond/MF/Unknown (rev-3 reason 2); D6 FLOW bid/ask features undefined before a root's `dataNotes` bid/ask start date (Q7, accepted 2026-11-02); D7 TOD on a root without a verified calendar: no current-day half-day flag, observed short days excluded from baselines, one Info line (N4 DECIDED). Rev 1's "Degrade" on RTH-only charts, on unknown early closes and on unconfirmed RTH windows is **withdrawn**. A new degraded mode needs Ettienne's approval.

Undefined outputs (NaN / false, one Info line, not a degrade) | # | Output | When | |—|—|—| | U1 | VPFUT overnight stats `ethHigh/ethLow/ethVolume` (L378) | `!hasOvernight` (N3 DECIDED) | | U2 | VPFUT `IsAsia()/IsEurope()` helpers (L244–245) | `!hasOvernight` (N3 DECIDED) | | U3 | DVP `rthShare` | `rthSplit == null` (already NaN on the generic base) |

Post-hoc calendar use without a verified table (correct, not degraded): `tradingDayCount` (L3228–3240) counts observed trading days (days that have bars) instead of NYSE days for roots without a table; `calendar = Observed`.

Own-contract availability (Ettienne 2026-11-06 12:13 ET, not a degraded mode and not a false result). When profile enforcement is on and the root has no roll facts, own-contract expiry/roll is `Unavailable`: `Step`/`Classify` still run, selected Macro rows and selected shared calendar rows (opex, VIX, rebalance, half day, Russell) still appear, and `isRollWeek` / `isQuarterlyExpiry` are null. They are not the equity-index dates and they are not false. `EventConfig.ownContract` off is `NotSelected` (also null, not false) and does not remove those shared rows. An explicit own-expiry/roll query throws S5. `NyseStandIn` does not become a verified calendar for `SessionsToExpiry`.

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7. Per-engine migration

Engine	Change	ES result (CME Globex template)	PR
CS host	L2462 typed; `autoTradingDay` and L1888–1889 → `tradingDayKey`; tick fallback → throw; `_NyseHolidays` for stocks, `p.calendar` otherwise; exchange map kept for ET conversions	identical	M1b, M4
CLOCK / VPFUT	Day key, day open, Globex open/close come from session data; RTH split per N1 (ES: today's constants, `rthSource=EquityIndexLegacy`); value window (09:30, 16:00] unchanged for ES; Asia/EU/NY windows per N5; holiday hooks per S4	identical	M4
CAL	ES keeps `_CmeHolidays` (both `UseCmeHolidayTable` flags, VPFUT L1627 and clock L81); NYSE for stocks; other roots `Observed` for post-hoc use and S4 throws for in-advance use until a verified table lands	identical	M8
ROLL	`_RollDetector` takes months/expiry/roll from the profile; F2; `gapTicks` stays 8 (Q4)	identical	M5
EVENT	Own `RollDate/ExpiryDate/SessionsToExpiry` from the profile when roll facts exist; separate `ownContract` / `sharedCalendar` controls (no inferred class mask). No	identical	M5

	facts: Step/Classify `Unavailable` (null, not false, not the ES quarter); explicit queries throw S5. NyseStandIn `SessionsToExpiry` still S4		
G0	Remove `tick = 0.25` (L357; throw if 0) **and update all 35 `new Gate0Engine` constructions in `tests/Gate0Tests/Program.cs` to set `tick = 0.25`** in the same PR (none sets it today); `"ES"` (L1204) → `p.root`; S3 coverage check; windows frozen	canary hash identical	M6a
FLOW	Tick-chunk boundary = `p.sessionEnd` (CME 16:00 CT = 17:00 ET, today's value); spot FX stays Unsupported (rev-3 reason 11); FX futures get their own session end	identical	M6b
DAILY / REGIME	Annualisation reads `p.sessionsPerYear` (252)	identical	M6c
LEVEL / AVWAP	Remove the 0.25 defaults (`_LevelStudy` L79, `_AvwapManager` L153); both hosts set them (L1410, L1976); AvwapTests already sets 0.25 (L17); `_AvwapManager` L195 already rejects ≤ 0	identical	M6d
RANGE / WEEKLY / PSTATS	`ringBars` **literals kept** (`_RangeProfile` L31 2880, `_ProfileStats` L47 1440; bar counts, not minutes; no derivation); WEEKLY `RthSessions` needs `rthSplit` (S7), else the user picks GlobexSessions; `_ProfileStats` 4-tick note in CFG	identical	M6e
PORT	`PortfolioCluster` returns `p.cluster`, same strings	identical	M6f
DVP	root from `p.root` (F1 done in M1a); roll table from the profile replaces `root == "ES"` (L213/230) and host loops (L119–121)	identical (post-#88 baseline)	M6g
REGGATE / `_Test6`	ES only (N2); windows frozen + S3; `tradingDay` hash text from an explicit `switch (ETradingDayKey) { case Globex18ET: return "Globex18ET"; … }` and `RollDetection` via an explicit switch with the same spellings (L62, `_Test6` L105–106); roll table from the profile (`_Test6` L100–102)	hash identical	M6h
SIZE / COST	`unitNotional` = entry × `p.bigPointValue`; cost floor = **new opt-in behaviour** (§9)	identical (off by default)	M7
MTF	`SymbolRoot` resolves via §4.3; doc fix (L28)	identical for ES	M1a
LEGACY	F3a/F3b only; others a one-line note	—	M1c

8. Presets: exchange facts only, no session times (sources read 2026-11-02)

8.1 Futures

Root	Tick	Big point value ($/1.0)	Tick $	Months	Expiry rule	Roll rule (Q5 I metho
ES	0.25	50	12.50	HMUZ	3rd Fri 09:30 ET SOQ, back to the NYSE business day	Mon before the 3r Fri ≥ 2022-04-01), days before (`_Ever` L299–303)
MES	0.25	5	1.25	HMUZ	as ES	as ES
NQ	0.25	20	5.00	HMUZ	as ES	as ES (**VERIFY** aga vendor splice)
MNQ	0.25	2	0.50	HMUZ	as ES	as NQ
CL	0.01	1000	10.00	all 12	3 business days before the 25th of the prior month (4 if the 25th is not a business day)	`BusinessDaysBefo` 2 business days be (**VERIFY** vs vendc
GC	0.10	100	10.00	active G J M Q V Z	third-last business day of the contract month	`BeforeFirstNotic` business day befo notice = last busin the prior month ('
ZN	0.015625	1000	15.625	HMUZ	7 business days before the last business day of the month	`BeforeFirstNotic` (**VERIFY**)
6E	0.00005	125,000	6.25	HMUZ (serials	2nd business day	`BusinessDaysBefo` 5 business days (a week) before expi

6L	0.00005	125,000	0.25	(serials listed)	before the 3rd Wed	(**Ettienne 2026-11** replaces “Mon of t week”)

Notes. No session, RTH, halt or Globex-time column: those come from the chart (R12). Rev 1’s RTH reference windows (CL 09:00–14:30, GC 08:20–13:30, ZN/6E 08:20–15:00 ET) and the ES 15:15–15:30 CT halt question are **dropped** (Q2 moot). `BigPointValue` for 6E is **CONFIRMED** 125000 (`01M6HTKV3SV1Q5QDXBFTCH8D3K.txt`); ZN is still **VERIFY** (not in the 13 runs); the cross-check (§4.3 step 8) refuses a mismatch. “Observed” calendars serve post-hoc use only; in-advance use throws (S4) until PR-M8 adds a sourced table. **Rev 2.4 (as merged, IP `InstrumentPresets` L215–237 @5c9ccd2).** CL/GC/ZN/6E use `ECalendar.NyseStandIn`: their expiry/roll rules count NYSE business days (`EventRules.IsNyseBusinessDay`, IP `InstrumentCalendars.BusinessDay` L439–446) until PR-M8 adds the CME calendars; the choice is explicit and the INSTR line says so (`calendar(NyseStandIn; calendar: NYSE stand-in (CME calendar pending M8))`, note text IP L437). NYSE and CME holiday schedules are not identical, so a computed date next to a holiday is VERIFY until PR-M8. No weekday fallback (R6). **MI-T4 fixture** `tests/InstrumentTests/fixtures/expiry-worked.csv` (11 rows: CL×4, GC×2, ZN×2, 6E×3): the `expiry` column was checked by the analyst on 2026-11-04 against CME notice chadv25-391, Barchart, CME rule 26101.G and the GC fact card; the `roll` column is **computed**: the preset rule applied by hand (e.g. 6EM24 expiry 2024-06-17, roll 2024-06-10), not vendor roll dates. The vendor-splice comparison stays VERIFY (PR-M5/M9).

8.2 Rule kinds (typed enums)

`EExpiryRule { ThirdFridaySoq, CL25thMinus3Bd, ThirdLastBd, LastBdMinus7Bd, ThirdWedMinus2Bd }`, `ERollRule { EquityIndexCustomary, BusinessDaysBeforeExpiry(n), BeforeFirstNotice(n), None }` (n is the row’s `rollDays`; the INSTR line prints `roll(BusinessDaysBeforeExpiry(5))`). `ECalendar { CmeEquityIndex, Nyse, NyseStandIn, Observed, None }`: `CmeEquityIndex`, `Nyse` and `NyseStandIn` give a business-day function, `Observed` and `None` give none, so a rule needing one refuses `calendar` (R6). ES/MES/NQ/MNQ use `ThirdFridaySoq` + `EquityIndexCustomary` = today’s `EventRules.RuleExpiryDate/RollDate`.

8.3 Non-futures

Class	Trading day	Calendar	Cost	Notes
Spot FX (12)	from the chart’s sessions (a 17:00 ET-ending template gives today’s ET+7h day)	Observed	Unknown (not costed until traded)	TickCount volume allowe DVP, G0, FLOW Unsuppor 3)
Crypto (20)	from the chart’s	Observed	Unknown	`sessionsPerYear` 365

	sessions			
Stock (2)	from the chart's sessions	Nyse (`_NyseHolidays`)	Unknown	FLOW and VPFUT Unsup (rev 3). SPY (**CONFIRMEI** `01M6HSDC62GCMK5ZANKB61P` `Stock`, ARCA, Build Volum TickCount (tagged `TickCo` no throw), Default sessio Fri 04:15–20:00 Exchange That template is day-only N1 its `rthSplit` is the wh 04:15–20:00 chart session extended hours included; only study needs a 09:30– template on the chart.

9. Sizing and costs (R8, R9): as the code is on `7dbdb1b`

1. q = 1% risk on the structural stop (CS-4 `sizeForRisk`, L2921), then `maxQty` (L2943) and the notional cap (L2945–2949) **reduce** q (Info when they bind). q never grows.
2. `willArm = m_portOn && bars == Bars` (L2957).
   - **Not willArm** (M-PORT off, or Data2..N): if q × stop × bigPointValue > 1% × equity the entry is **skipped**, Info "trade skipped" (L2958–2968).
   - **willArm** (portfolio Data1): the order goes to `portfolioArm` (L2970) and `_PortMM` applies deny-not-reduce. M-INSTR changes nothing here.
3. **Cost floor** = **new opt-in behaviour (PR-M7)**: `InstrumentConfig.applyCostFloor = false` by default (no input). When on, the stop is widened to `max(structural, costFloorTicks × tick)` before step 2, so the existing skip/deny paths act on the widened stop. Rev-1's claim that main already does this is withdrawn (`costFloorTicks` exists only in comments, `_Gate0` L1185, `_TestGate0` L145).
4. `costFloorTicks` in integer cents: `tickValueCents = tick × bigPointValue × 100`, which must be within 1e-6 of an integer (else throw `mismatch:tickValue`); `costFloorTicks = ceil(roundTripCents × 10 / tickValueCents)` in `long` arithmetic, `(a + b - 1) / b`. ES: 1700 × 10 / 1250 → 14. MES: 250 × 10 / 125 = 20 exactly (no float path can give 21). 4a. **As merged (M2/M3, IP `CostModel.FloorTicks` L379–389, Build L778–779 @5c9ccd2).** The floor is computed **only when a preset matched and the config is valid**, using the preset's tick and bigPointValue; otherwise `costFloorTicks = -1` (`floor(Unknown)`). A known cost with a negative cents value refuses `unknown-field:cost`; the non-integer tickValue is `mismatch:tickValue` (enforced in M3). Rationale: with M3's root-only keys (§5.3), a stock ticker `ES` must not get a floor from the ES future's cost. **PR-M4 (D1 DECIDED):** costs are keyed by class + root (§5.3), so the cost itself can no longer cross classes; the preset gate stays. PR-M4 also prints why the floor is Unknown: `floor(Unknown src=no-preset|no-cost|cost-class-mismatch|config-invalid)` (§6.1); `cost-class-mismatch` = a row exists for the root under another class and was not used.
5. Costs: `p.cost` from `InstrumentConfig.costs` (cents), keyed by **asset class** + **root** from PR-M4 (§5.3, D1 DECIDED 2026-11-04; M3 keys by root only). DECIDED: `Future:ES` 1700, `Future:MES` 250. Other roots: the method is

DECIDED (Q3, IBKR fee per side ×2 + one tick slippage per side ×2), but the per-root IBKR fees are a **data input from Ettienne**; until given, cost = Unknown and cost users throw (S5).

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10. Tests (MC-free; new `tests/InstrumentTests` in `PLStudies.Tests.sln`, `CheckForOverflowUnderflow=true`, real sources linked via the `gen.sh` region pattern)

ID	Test
MI-T1	(Not a test.) `ClassOf` is the exhaustive host switch of §4.2, checked by the PLEditor compile in PR-M3. The MC-free core is tested on `EAssetClass` only. **Result 2026-11-04 (Ettienne, `5c9ccd2`): Compile All passed** after the one-time import of `_InstrumentProfile.Function.CS`; `eMTPA_SF_Root`, `eMTPA_MCSC_USD` compile.
MI-T2	Root resolver (§4.3): @ES, @ES#C, ESZ24, ESZ4, **ESZ2024 → ES, 'ES 12-24' → ES, ESZ24:XCME → ES, CLZ2026 → CL, @6E → 6E, 6EZ26 → 6E, EUR/USD → EURUSD, BRK.B → BRK (M1a; class-aware stock handling in the M2 resolver keeps BRK.B), Z4 → Z**, MESZ26 → MES, MESH27 → MES, **6E vs 6B differ**, @MNQ, GCQ26, ZNH27, **EUR/USD vs EUR/JPY differ**, BTCUSD, AAPL; M2: / removed for Spot FX only; M2 root source: MC root primary, parser fallback only on `(UnsupportedFields & eMTPA_SF_Root) ≠ 0` (incl. -1) or empty root, `!SameRoot` → Warning + MC wins; **`_RootDump` fixtures (§4.4.3):** (@ES, ES, 0) → ES/mc, (@CL, CL, 0) → CL/mc, (@6E, 6E, 0) → 6E/mc, (SPY, SPY, 0) → SPY/mc, (@6E, "", 0) → 6E/parsed, (@ES, ES, 1) → ES/parsed, (@ES, ES, -1) → ES/parsed, (@ES, ES, 512) → ES/mc; SymbolExpiry 2038-01-01 / 1900-01-01 → placeholder (calendar), 2026-12-18 → used;"ES" as a stock gets no preset; empty root rejected; `RootError("", "")` is an error (F1). Preset cross-check: tick/bigPointValue mismatch throws `mismatch:tick` / `mismatch:bigPointValue`. MC-

MI-T3	internal cross-check on the four `_RootDump` rows (§4.4.3) passes: 0.01 × 100, 1E-05 × 99999.999999999985 (within 1e-9), BPV 50/1000/125000/1 equal; perturbed rows (struct PriceScale 0.1 vs 100; BPV 50 vs 5) throw `mismatch:mcPriceScale` / `mismatch:mcBigPointValue` with the message naming both values.
MI-T4	Expiry rules vs CME calendar dates per root 2016–2027 (fixture CSV with the source URL); ES rows = `EventRules.RuleExpiryDate`. **As merged (IT T4 L196–242):** `fixtures/expiry-worked.csv` (expiry analyst-checked, roll computed from the preset rule, §8.1); NyseStandIn on CL/GC/ZN/6E; 6E `BusinessDaysBeforeExpiry(5)` (L219); no weekday fallback (an Observed/None calendar refuses `calendar`, L228); `months` refusal (L241–242).
MI-T5	Roll rules: the ES set reproduces `RollDatesDigest2000To2035 = "fdd63bd8f179a8d4"` (`_RegimeGate` L60).
MI-T6	`SeriesWindow` generalised: `tests/RollTests` unchanged; CL/GC monthly windows.
MI-T7	Trading day from sessions: CME Globex template (Sun 17:00 → Mon 16:00 CT, and the ET equivalent) gives `GlobexTradingDayEt` for every 1-min close over 2016–2024 incl. DST weeks; a two-session day (GLOBEX 17:00–15:15 + 15:30–16:30, the 9-entry list of `01M6HSDBFRQ9ZZFA4NBXZ8QZXC.txt`) shares one key; FX 17:00 ET template = ET+7h; 00:00 UTC crypto = UTC date; a day-only template keys by calendar date; a bar outside every session throws.
MI-T8	Derived fields: `dayOpen`, `sessionEnd`, `hasOvernight`, `rthSplit` (day-only template → chart session; Globex template + ES → `EquityIndexLegacy` equal to `_CmeSessionClock` defaults field by field; other 24 h → null).
MI-T9	Cost floor in integer cents: ES 14, **MES 20**, a non-integer tickValueCents throws.
MI-T10	Sizing: caps reduce q; the off-portfolio skip logs Info; floor off by default changes nothing; q never

	grows.
MI-T11	Exception text per reason code (§6.3), incl. `sessions:*`. (The T11 list, IT L440, does not include `months`; `months` is tested in T4.)
MI-T12	INSTR/CFG line formats pinned; culture-invariant (run under de-DE). **As merged:** full ES INSTR line (IT L537–539) and RUN suffix format (L544) pinned (§6.1); M3 ordering, enforced/log-only and Guard tests IT L596–630.
MI-T13	`profileHash` stable and outside every `configHash`. **As merged (IT L446–476):** ES `020deca78d6392b7`, 6E `8e8b3e9b0e230e12` (L472–473), full 6E line (L475). MC (2026-11-04): real 6E chart `55576df6a8a99c05`, reproduced byte-for-byte on .NET 8 (runtime-stable). **PR-M4 (D2):** add the tz-invariance tests and the expected pins of §6.1 / §11.1 item 9. `MonthMask.Of` bad-letter test L631–640.
MI-T14	Lint (in `tests/RepoLintTests`, PR-M1d) **scoped to MC enum types** (`ESymbolCategory`, `RequestTimeZone`, `EResolution`, `BuildVolumeOn`, `MTPA_MCSymbolCurrency`, `MTPA_MCSymbolCategories`, `MTPA_SymbolFields`, `DayOfWeek` from `SessionObject`): fails on `.ToString()` of those values feeding a comparison or a parse, on `Enum.Parse`/`TryParse` with those types, and on `System.Reflection` / `GetProperty(`. Exact allow-list (file + line pattern) as of M4 @d3c0c1a: `_CustomSignal` inputSnapshot reflection, `_MarketBreadth.Strategy.CS` `using System.Reflection;`, `barZoneLabel` (log), the `SeriesKeyWarning` text (log). **`_TestPortMM` L90 removed with the file in PR-M4 (A2)** — a stale allow-list entry fails the "no stale entries" check. Repo-enum parses (`_EconomicCalendar`, `_StructuralTargetPatterns`, `_EventCalendar`, `_PortMM`) are out of scope.
MI-T15	ES bit-identity per PR: `_TestGate0` canary hash and the DVP D9a summary (**post-#88 baseline**) unchanged; ES fixtures in `tests/*` unchanged; requires **Ettienne's MC runs for every PR**, numbers

		pasted into the PR.
MI-T16		Session ends from the list (R14): the 9-entry GLOBEX list of `01M6HSDBFRQ9ZZFA4NBXZ8QZXC.txt` with 1-min bars and `LastBarInSession` false at 15:15 (true only at 16:00, as dumped) gives session-end events at **15:15 and 16:00** each weekday and trading-day rollover after 16:00; a duplicate CalcBar at the same 16:00 time (`01M6HSDBFH2D6H21PMYXYKFJHV.txt`) resets once; a session-reset engine fixture (VPFUT/AVWAP/FLOW reset hook) sees exactly two resets per GLOBEX day and one per Default day; a list with no bar at its `EndTime` (data stops early) ends the session on the next out-of-session bar or on `LastBarInSession`.

State on `d3c0c1a` (rev 2.5): `tests/InstrumentTests/run.sh` → `InstrumentTests: 756 passed, 0 failed` on the M4 head (box, .NET 8, 2026-11-04 ET); full check **23 suites, 4528 passed, 0 failed**. Pins: 6E `55576df6a8a99c05`, ES `6cff053d1b215324`, AAPL `41886799d411dd0d`.

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11. PR list (each small; each targets `main`, opened after the previous one merged; no stacked PRs)

Status on `d3c0c1a` (rev 2.5): M0 **#89** merged, M1a **#90** merged, M1b **#91** merged, M1c **#92** merged, M1e **#93** merged, M1d **#95** merged, M2 **#96** merged, M3 **#97** merged, **M4 #98 merged** (`d3c0c1a`, final head `2418052`). **Extra, outside this list: #94** `_ExitHalfRiskHalfRoll` exit-only (half off at +1R, trail the rest; exit-only invariant; `OrderExit.Total`), merged (`ed3e4a5`); it changes the legacy study only. Next: **PR-M5**.

#	PR	Content	Behaviour change
—	**#88**	Land #86 + #87 (existing)	—
0	PR-M0 **#89 merged**	`_InstrDump.Strategy.CS` (§4.4, derives from `_CustomSignal`) + the `LogDirectory()` accessor in `_CustomSignal` (§4.4.1) + csproj entry + import note; `MCInstrDump_*.txt` output committed as fixtures after Ettienne's runs	none
1	PR-M1a **#90 merged**	F1: `_MtfMath.SymbolRoot` (§4.3 steps 1–2: leading digits + letter run, month drop, "/" removed; L109–119 on `6853398` before #90, L120–135 after #90), doc L28, `RootError` empty = error, roll inheritance and same-root checks	digit-led roots / FX only
	PR-M1b	F2 (isFuturesCategory at	

2	**#91 merged**	L1069/L1866/L1882), F4 (tick throw), L2462 typed	FutureRolover, bad tick
3	PR-M1c **#92 merged**	F3a + F3b in `_ExitHalfRiskHalfRoll` only	legacy study on
4	PR-M1e **#93 merged**	Retire `_InstrProbe` + `_InstrProbeCore` + `tests/InstrProbeTests` + csproj L521–522 + sln entry	none
5	PR-M1d **#95 merged**	MI-T14 lint + allow-list	none
6	PR-M2 **#96 merged**	`_InstrumentProfile.Function.CS` core (enums, resolver, presets, `InstrumentConfig`, `CostModel`, exceptions, session derivations on structs), `<Compile Include>` + import order, `tests/InstrumentTests` + sln, rev-3 “superseded for C1–C9” banner (Q10)	none (not wire
7	PR-M3 **#97 merged**	Wire `buildProfiles()` + `SessionReader` (StartCalc read + first-CalcBar re-check), `ClassOf` switch, INSTR line, cross-checks, `profileHash` on RUN	log only (+ `mismatch:*` and `sessions:chang` throw, §6.2)
8	PR-M4 **#98 merged** (`d3c0c1a`, head `2418052`)	Enforce every §6.3 refusal globally (A1); D1 class:root cost keys; D2 exchange-time profileHash via `exchangeZoneIdOf` token match; strict tick / FALLBACK_TICK removed; floor `src=…`; PORT publish line deferred to first CalcBar; `_TestPortMM` deleted (A2); DST tests; Start marks started only after Guard(Build). Scope detail: §11.1	refusals → Erro + throw on eve stream for all `_CustomSignal` studies; same instrument hashes equal across chart/PT zones
9	PR-M5	Expiry/roll from the profile: `_RollDetector`, `EventRules`, host quarterly loops (`_DvpEngineTest`, `_Test6`). Rev 2.5.1: separate `ownContract` / `sharedCalendar`; no roll facts → `Unavailable` on Step/Classify (not the ES quarter, not false); explicit queries throw S5	non-ES only; ES identical
10	PR-M6a	G0: tick default removed + **Gate0Tests 35 constructions set tick**; root; S3	none for ES
11	PR-M6b	FLOW: chunk boundary = `sessionEnd`	non-ES only
12	PR-M6c	DAILY/REGIME `sessionsPerYear`	non-futures on
13	PR-M6d	LEVEL/AVWAP tick defaults removed	none
14	PR-M6e	RANGE/WEEKLY/PSTATS: S7 for `RthSessions`; CFG note; ringBars literal	none for ES
15	PR-M6f	PORT cluster from profile	none
16	PR-M6g	DVP roll table from profile	none for ES
17	PR-M6h	REGGATE: ES only (N2), explicit hash switches, S3	hash identical
			none unless

18	PR-M7	Cost floor (opt-in, new) + costs in cents	enabled
19	PR-M8	Calendars for other roots, only when sourced (ES table unchanged)	non-ES only
later	PR-M9	Self-calibrating roll gap (Q4); ES stays 8 until it reproduces `tests/RollTests` and the m-event splices; `tools/m-event/out/roll_splices_2016_2024.csv` is read-only	none for ES

11.1 PR-M4 scope — DONE (rev 2.5; merged #98 @d3c0c1a)

1. **Log-only → enforced (A1, Ettienne 2026-11-04):** *"Keep the strict version: stop any strategy on any bad stream."* Every refusal of §6.3 (incl. `class`, `unknown-field:*`, `tick`, `currency`, `calendar`, `months`, `sessions:*`, `tz:bar`, `tz:exchange`, `mismatch:*`) is Error + throw **globally at StartCalc on every data stream for all `_CustomSignal`-based studies**, not only for engines that use the field. The `INSTR ds(n) log-only…` line and the null-profile / `profileHash(…=none)` path are gone. `InstrumentStreams.Start` sets `m_started` / `m_pending` only after `Guard(Build)` returns. **Refused:** Unsupported/unmapped class; non-USD; unresolved bar/exchange zone (unmapped exchange → `tz:exchange`); PriceScale ≤ 0 / tick ≤ 0; missing or changed sessions; failed typed/session reads; invalid `InstrumentConfig` (incl. old root-only cost keys → `unknown-field:config`); `mismatch:*`; `calendar` / `months` when rules need them. **Still runs:** futures with no preset (cost/floor Unknown); USD stocks on NYSE/NASDAQ/ARCA.
2. **ClassOf:** an unmapped `ESymbolCategory` in `buildProfiles()` is Error + throw (same as `classOfData1`). **DONE.**
3. **FALLBACK_TICK audit (R6):** `FALLBACK_TICK = 0.01` removed from `_VolumeProfileBase.Strategy.CS` and `_VpConfluenceEngine.Function.CS`; tick comes from the Data1 profile (or the strict host tick when M-INSTR is off). **DONE.**
4. **`tickFromMinMove` audit:** the `PriceScale == 0 → MinMove` path and `Math.Abs` of a negative ratio are gone; a tick that cannot be resolved is Error + throw. Host `tickSize()` prefers the Data1 profile tick. **DONE.**
5. **Floor-Unknown reason** on INSTR: `floor(Unknown src=no-preset|no-cost|cost-class-mismatch|config-invalid)`. **DONE.**
6. **Portfolio Trader test, before M4 starts: DONE 2026-11-04** (Ettienne, `5c9ccd2`): one PT log for @6E, @ES, AAPL (§4.4.2).
7. **`eMTPA_SF_Root`: DONE 2026-11-04**: MI-T1 compile passed, `rootSrc=mc` on 6E/ES/AAPL (§4.1); the MC 6E hash `55576df6a8a99c05` reproduces on .NET 8 (`8e8b3e9b0e230e12` is the synthetic-list pin).
8. **Cost keying by asset class + root (D1): DONE.** `CostKey(EAssetClass, root)` = `<Class>:<ROOT>`; defaults `Future:ES` 1700, `Future:MES` 250; `Validate` rejects root-only and unknown-class keys (`unknown-field:config`); a same-root row of another class is not used (`cost(Unknown)`, `floor(Unknown src=cost-class-mismatch)`, one Warning).
9. **profileHash in exchange time (D2): DONE.** `Canon` hashes sessions and rth in the exchange zone via `exchangeZoneIdOf` (token match, §6.1); refusals `tz:bar` / `tz:exchange`; INSTR adds `xsessions(…)`; engine-facing fields stay in bar time. Pins confirmed in IT: 6E Default `55576df6a8a99c05`, ES Default `6cff053d1b215324`, AAPL `41886799d411dd0d`. DST tests (summer, spring-forward, fall-back) keep the 6E pin. **Still OPEN:** GMT chart VERIFY + live check of composite exchange strings (**A3**, §12.4).

10. **PORT publish line timestamp: DONE.** `PORT publish on: …` deferred from StartCalc to the first CalcBar (calc thread: bar date + zone known).
11. **`_TestPortMM` deleted (A2, Ettienne 2026-11-04).** File `git rm`'d; MI-T14 allow-list entry removed; `_TestPortArm` remains (header notes the deletion). **Open:** live-check-sheet Check 7 MM-side checks need a replacement MM test signal before that check can run (§12.4).

Every PR lists the R1–R14 rules it touches, the VERIFY items it closes, its csproj/sln changes and the PowerLanguage Editor imports, and re-reads its line cites on current main.

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12. Questions

12.1 DECIDED

- **Q1 / Q2 Sessions and RTH windows: DECIDED by Ettienne 2026-11-02.** Verbatim: *"The trading hours in questions 1 and 2 should be driven from Multicharts - it has session information you can get and use. This info is setup from the chart and just work with that. Each instrument also has a 'Default' setting which should be correct and used the traders in general."* Applied as R12 and §4.5. Q2 (RTH windows for CL/GC/ZN/6E) is moot: no RTH windows in presets.
- **Q3 Costs for other roots: DECIDED** (Ettienne 2026-11-02 accepted the recommendation): IBKR fees + one tick slippage, as ES/MES. **Data input outstanding:** IBKR per-side fees for NQ, MNQ, CL, GC, ZN, 6E; until given those roots have cost Unknown.
- **Q4 Roll-gap threshold: DECIDED** (Ettienne 2026-11-02 accepted the recommendation): self-calibrating in PR-M9, only after it reproduces RollTests and the m-event splices; ES keeps 8 until then.
- **Q5 Roll rules for CL/GC/ZN/6E: DECIDED** (Ettienne 2026-11-02 accepted the recommendation): rule-based table per §8.1, confirmed against the vendor's continuous splice; series detection fallback.
- **Q6 Retire `_InstrProbe`: DECIDED** (Ettienne 2026-11-02 accepted the recommendation): PR-M1e.
- **Q7 Data-availability notes: DECIDED** (Ettienne 2026-11-02 accepted the recommendation): `dataNotes`, logged on INSTR; FLOW bid/ask features undefined before the date (approved degraded mode D6, §6.4). **Data input outstanding:** dxFeed tick bid/ask start dates for @ES and @MES.
- **Q8 Non-USD: DECIDED** (Ettienne 2026-11-02 accepted the recommendation): USD only; non-USD throws `currency`.
- **Q9 Stop floor: DECIDED** (Ettienne 2026-11-02 accepted the recommendation): cost-derived, integer cents (§9.4), opt-in (PR-M7).
- **Q10 Supersede rev 3: DECIDED** (Ettienne 2026-11-02 accepted the recommendation): banner in PR-M2.

12.2 N1–N6: DECIDED (Ettienne 2026-11-03 09:47 ET: *"1 to 5 is fine, on 6 keep that dump in the same directory as the created standard log file"*)

- **N1 RTH split on a 24-hour chart: DECIDED as recommended (B).** ES/MES/NQ/MNQ on a 24-hour chart keep today's equity-index split in code (`rth src=EquityIndexLegacy`, ES bit-identical); other roots on a 24-hour chart throw S7 for RTH-dependent engines; on a day-only chart the chart session is the RTH. Option A (a second RTH-template stream) is not used because of MC-1439 stalls under Realtime-History Matching.

- **N2 A2-63 on MES: DECIDED as recommended.** REGGATE accepts ES only; MES would be a new trial with its own pre-registration.
- **N3 Overnight-only outputs without an overnight segment: DECIDED as recommended.** NaN/false with one Info line (U1–U2), not a throw.
- **N4 TOD half-day flag without a verified calendar: DECIDED as recommended.** Approved degraded mode D7 (§6.4).
- **N5 Asia/Europe/NY clock windows: DECIDED as recommended.** Kept as instrument-independent code defaults in ET (VPFUT L52–63, `_CmeSessionClock` L48–61), clipped to the chart's sessions; not presets.
- **N6 `_InstrDump` output: DECIDED.** Same directory as the standard MCSignal log, via the host's resolver (`_CustomSignal` `LogFileSingleton` L71–91, exposed through a new `LogDirectory()` accessor); file `MCInstrDump_<symbol>_<yyyyMMdd-HHmmss>-<pid>.txt` (§4.4.1).

12.3 Data inputs still pending (do not block PR-M0 to PR-M3)

- Q3: IBKR per-side fees for NQ, MNQ, CL, GC, ZN, 6E (until given: cost Unknown, S5).
- Q7: dxFeed tick bid/ask start dates for @ES and @MES.
- PR-M0 OPEN items (§4.4.2): runs not yet made: @ZN, one FX pair, one crypto symbol. The Portfolio Trader per-symbol item is **CLOSED** (PT test passed 2026-11-04, §4.4.2, §11.1 item 6).
- **MI-T1 / MI-T15 (Ettienne): DONE 2026-11-04 on `5c9ccd2`.** Compile All passed (one-time import of `_InstrumentProfile.Function.CS`; `eMTPA_SF_Root`, `eMTPA_MCSC_USD` compile). 6E chart (Exchange tz, Default, BVO TickCount): `tick(5E-05) bpv(125000) tickValue(6.25)`, NyseStandIn note, `roll(BusinessDaysBeforeExpiry(5))`, profileHash `55576df6a8a99c05`, reproduced byte-for-byte on .NET 8; the pins `020deca78d6392b7` / `8e8b3e9b0e230e12` are for the synthetic 9-session list and are not expected in MC.

12.4 Decisions (rev 2.5)

- **D1 Cost keying: DECIDED by Ettienne, 2026-11-04: key costs by asset class + root** (e.g. `Future:ES`), so a configured futures cost never applies to a stock with the same root. Rule, key format, default mapping (`Future:ES` 1700, `Future:MES` 250), class-mismatch handling and the configHash check: §5.3; floor reasons: §9 4a; implementation: **DONE in PR-M4** (§11.1 item 8). M3 (`5c9ccd2`) keyed by root only; M4 keys by `CostKey(class, root)`. Options considered: (a) keep root-only plus the preset gate; (b) class + root (recommended, chosen).
- **D2 profileHash time zone: DECIDED by Ettienne, 2026-11-04: compute profileHash from the sessions expressed in exchange time**, so the same instrument hashes the same whatever the chart / Portfolio Trader time zone. Evidence: @6E `55576df6a8a99c05` (chart, Exchange tz, Sun 17:00→Mon 16:00 … Central) vs `9523b25025f9ed5b` (PT, Local, 18:00→17:00 ET); both reproduced on the box from the canon. Zone source, refusals (`tz:bar`, `tz:exchange`), conversion, INSTR `xsessions(…)`, rth handling and expected pins: §6.1; implementation and tests: **DONE in PR-M4** (§11.1 item 9). M3 (`5c9ccd2`) hashed bar-time sessions; M4 hashes exchange-time sessions via `exchangeZoneIdOf`.
- **A1 Enforcement scope: DECIDED by Ettienne, 2026-11-04 (own words):** *"Keep the strict version: stop any strategy on any bad stream."* M4 enforces every §6.3 refusal globally at StartCalc on every data stream for all `_CustomSignal`-based studies — not only for engines that use the field. Refused vs still-runs list: §11.1 item 1 / §6.2.
- **A2 `_TestPortMM`: DECIDED by Ettienne, 2026-11-04 — delete it.** `_TestPortMM.Signal.CS` removed in PR-M4; its MI-T14 allow-list entry

removed; `_TestPortArm` remains (comments updated). **Open item:** live-check-sheet Check 7 MM-side checks (P5b L1–L9) need a replacement MM test signal before that check can run (`docs/engines-manual.md` notes the removal).

- **A3 Composite exchange names + GMT VERIFY: OPEN.** `exchangeZoneIdOf` requires exact tokens; composite / feed names (NYSEARCA, NASDAQGS, CMEGLOBEX, AMEX, BATS, CFE, and similar) are deliberately unmapped → `tz:exchange` until a live check of real `Bars.Info.Exchange` strings on dxFeed shows what the feed reports. Separately, one MC run on a **GMT** chart is still VERIFY (does MC shift GMT sessions across DST? the box conversion uses the first bar's week).
- **Own-contract controls: DECIDED by Ettienne, 2026-11-06 12:13 ET (own words):** *"I approve separate expiry/roll controls in M5, preserving other calendar events, explicitly marking unsupported contract-expiry status unavailable, and rejecting explicit unsupported expiry/roll queries."* Applied in PR-M5 (§6.4 own-contract availability, §7 EVENT, rev 2.5.1). Shared calendar rows stay independently selectable. No new `[Input]`, no per-asset-class event mask, no NyseStandIn session count, no cost-arithmetic change.
- **Accepted, recorded:** `MonthMask.Of` throws `ArgumentException` on a bad month letter **without** an Error line (exception to "Error before every throw"). It is called only on static preset data, the doc comment says so, and IT L631–640 tests it; a bad letter fails every test run before any MC run. No change.

12.5 Code vs spec divergences found for rev 2.4 (on `5c9ccd2`; several closed by PR-M4 @d3c0c1a — see V6, V7, V8, V15 below)

#	Where	Spec said (rev 2.3)	Code does
V1	§5.1 `calendar`	`{CmeEquityIndex, Nyse, Observed, None}`	adds `NyseStandIn` (IP L33)
V2	§8.1 CL/GC/ZN/6E	calendar Observed; 6E roll "Mon of the expiry week"	NyseStandIn; 6E `BusinessDaysBeforeExpiry(5)`
V3	§4.3 step 8	6E tick `5.0000000000000007e-5`	raw `5.0000000000000009E-05` (R, Framework); profile `5E-05`
V4	§5.1 `dataNotes`	list of (from, what)	one string per root
V5	§6.1 example	no `rootSrc`, no `dataNotes`, `sessionEnd(16:00)`, n=1	`root(ES rootSrc=mc)`, optional `dataNotes(…)`; GLOBEX 9-entry list ends 16:30
V6	R13/R6 "Error before every throw"	every refusal Error + throw	M3 log-only refusals wrote only a Warning; enforced ones wrote the Error
V7	R6	no silent fallbacks	`FALLBACK_TICK` in VP/VpConfluence; `tickFromMinMove` PriceScale==0 → MinMove and `Math.Abs`

V8	§4.2	unmapped category throws	`buildProfiles` turned it into a log-only Warning (`classOfData1` still threw)
V9	§5.1/§9 floor	floor from cost	only with a matched preset and a valid config; cost still shown without a preset
V10	R6 `calendar`	rules without a calendar throw	Build refusal is log-only in M3; reachable only via an override row (presets all have a calendar; `Validate` does not reject it)
V11	§6.2	“null profile” on refusal	a first-bar session refusal keeps the StartCalc profile (INSTR still printed, `sessions(n=0)` when none were read)
V12	MI-T11	every reason code	T11 list omits `months` (covered in T4)
V13	§4.1	`eMTPA_SF_Root = 1` stated as fact	mirror-only; not in the PG or a dump
V14	§5.3 costs	class + root keys (D1, 2026-11-04)	root-only keys (`ES`, `MES`); a stock `ES` shows the future’s cost
V15	§6.1 profileHash	sessions/rth hashed in exchange time (D2)	M3 hashed in bar time; @6E `55576df6a8a99c05` (Exchange) vs `9523b25025f9ed5b` (PT Local)

Appendix A. Review responses (Coder review of rev 1, 37 items; each re-checked on `7dbdb1b`)

#	Item	Response	Where
1	Lint fails on main; needs allow-list, MC enums only	Accepted. Scoped to MC enums with an exact allow-list; remaining MC-enum string uses listed as findings	§3.3, MI-T14, PR-M1d
	Removing `_InstrProbe` breaks	Accepted. Own PR with the suite, csproj	

2	InstrProbeTests	L521–522 and sln; M1 does not depend on the M0 dump	PR-M1e
3	Removing G0 tick default breaks Gate0Tests	Accepted, with a correction: 35 `new Gate0Engine` constructions and **none** sets the engine tick (L168/184/191 are constructions); all 35 updated in PR-M6a	§7, PR-M6a
4	ringBars not ES-unchanged if derived	Accepted. Literals kept (2880; `_ProfileStats` 1440); they are bar counts	rows 55, 65; §7
5	§9.1 vs willArm	Accepted. §9 restates the code; floor is new opt-in	§9
6	Throw vs degrade conflicts	Accepted. One rule (R13), every case listed; rev-1 degrades withdrawn	R13, §6.4
7	Row 13 cite	Accepted: L1840–1842	row 13
8	Row 23 missing `MultiChartsProfile=0`	Accepted	row 23
9	Row 24 value window (09:30,16:00]	Accepted	row 24
10	Rows 26/28 VPFUT own table	Accepted	rows 26, 28
11	Counts	Accepted. One area per row, totals 70	§3.1
12	L2621 NYMEX/COMEX → Eastern	Accepted as a finding; map kept only for ET conversion; fixed in M4 if dump A3 shows Central	row 4, §4.4 A3, §4.5
13	Host quarterly tables	Accepted	row 64, PR-M5
14	L1888 Eastern	Accepted; removed by session-derived day key	row 63, §4.5

15	`_ProfileStats` 4-tick; NQ $2.50	Accepted	rows 66, 67
16	`SymbolRoot` doc	Accepted	row 68, PR-M1a
17	F1 wider	Accepted; MI-T2 adds 6E/6B and EUR/USD vs EUR/JPY	§3.2, MI-T2
18	F2 also L1069/L1866	Accepted	§3.2, PR-M1b
19	F3 description + second bug	Accepted	rows 60, 70; PR-M1c
20	F4	Accepted	PR-M1b
21	`Bars.Sessions` compiles	Accepted; StartCalc population, zone and per-series sessions CONFIRMED by the PR-M0 dump (rev 2.2); read at StartCalc, re-checked on the first bar	§4.1, §4.4 A2–A4, §4.5
22	`StrategyCurrencyCode` on plain signal	Accepted; moved to section B	§4.4 B2
23	Drop Description/PointValue/Resolution.Size from dump	Accepted	§4.4 A1
24	Dump3 hygiene	Accepted; cited as box-only run output; M0 commits typed output	header, §4.4
25	Rename to `bigPointValue`	Accepted	§5.1
26	Split M1	Accepted (a, b, c, d + e for the probe)	§11
27	Sequential PRs on main	Accepted; #88 first, no stacking	header, §11
28	M2 build plumbing	Accepted	PR-M2, §10 header
29	MI-T1 cannot be a test	Accepted; host exhaustive switch	§4.2, MI-T1
30	MI-T15 baseline + MC runs	Accepted; post-#88 baseline, Ettienne's runs	MI-T15

		per PR	
31	Banner in M2	Accepted	PR-M2
32	R3 PriceAction preset	Accepted; preset clause dropped, “PriceAction unchanged” kept	R3
33	R4 wording	Accepted: “no new [Input]s”	R4
34	REGGATE Q8 cross-ref; hash switch	Accepted; cross-ref fixed, MES as new trial raised as N2, explicit switches	§7, N2, PR-M6h
35	Cost floor exact integers	Accepted; integer cents, MES = 20 test	§9.4, MI-T9
36	FLOW boundary for FX	Changed: boundary = the chart’s session end (no per-class rule); spot FX is already Unsupported for FLOW (rev-3 reason 11); 6E/FX futures use their own session end	§7, PR-M6b
37	Q4: ES stays 8; CSV read-only	Accepted	Q4, PR-M9

Rejected outright: none. **Partly changed:** item 3 (no Gate0 test sets the tick, so all 35 are updated) and item 36 (session end instead of a per-class rule).